Block practice · Homeostasis: skin, kidney and liver

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Identify urinary system and nephron structures

Explain homeostasis and negative feedback

Explain how insulin and glucagon control blood glucose

Explain how skin blood flow and sweating control body temperature

Explain the liver's roles in deamination and breaking down drugs

Explain ultrafiltration and selective reabsorption in the nephron

Explain how ADH changes urine volume and concentration

Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

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Practice workbook · Identify urinary system and nephron structures

Naming, labelling, drawing or matching kidney, ureter, bladder, urethra and nephron parts, including assigning functions to structures in a table.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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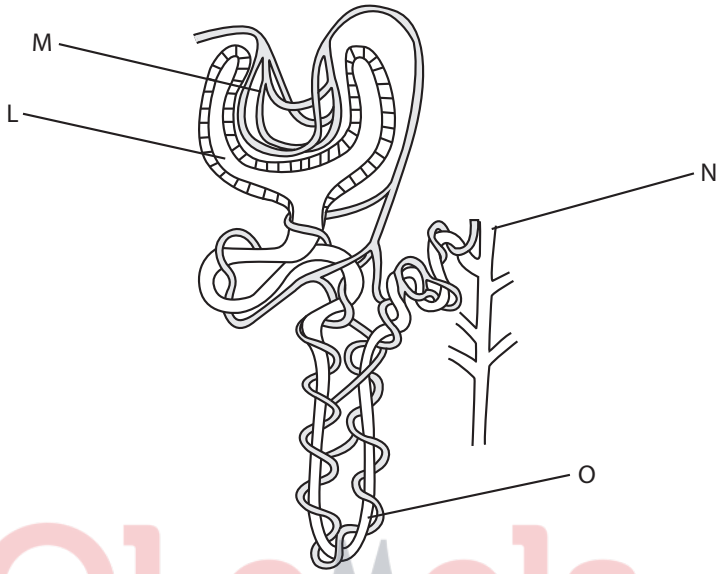
Type ID: HBI-NAME-RENAL-STRUCTURES

1. Question 6 · part a · 4 marks

November 2023 Paper 1

Worked-example retry · Attempt part a only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O.

(4)

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L.

(3)

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1. Question 6 · part a · 4 marks · Working and answer

Working and answer

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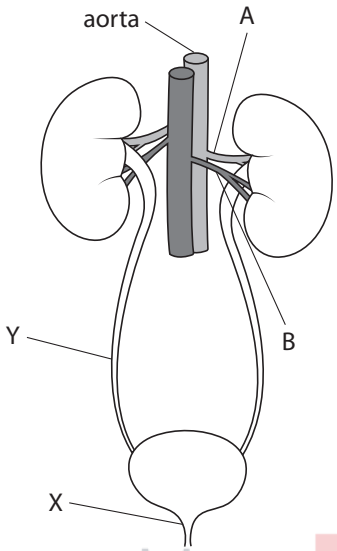
2. Question 8 · part a.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

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8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y. **(2)**

structure X .....

structure Y .....

(ii) Describe the function of the aorta. **(2)**

.....

.....

.....

.....

(iii) Compare the composition of the blood in vessels A and B. **(2)**

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2. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 9 · part b · 4 marks

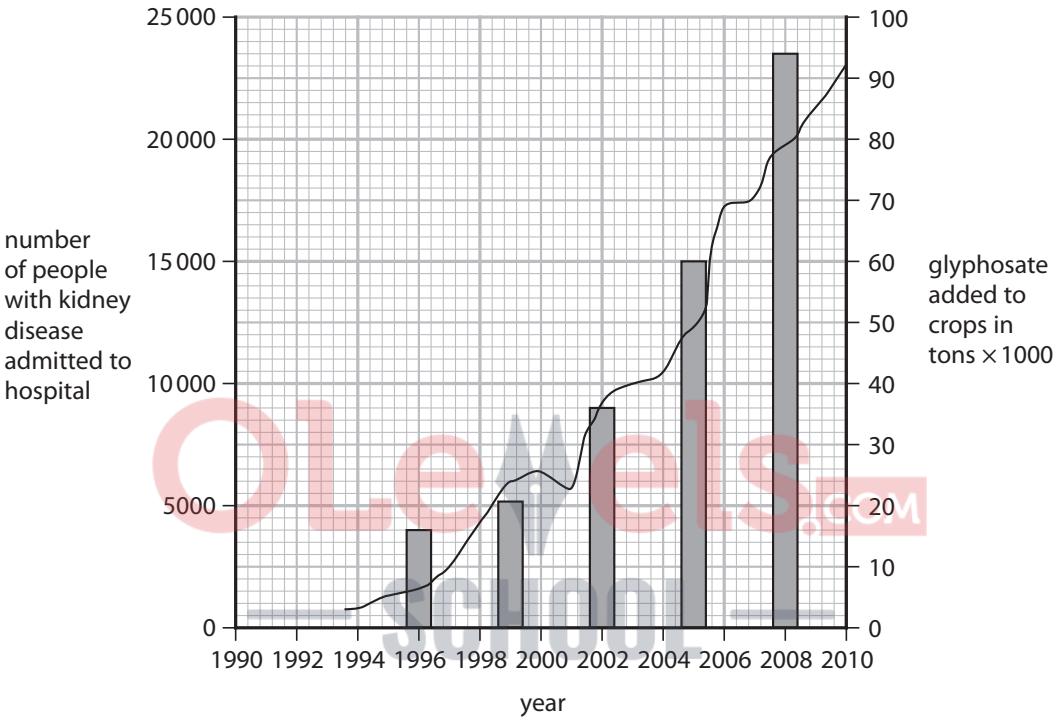
November 2020 Paper 1R

Further practice · Attempt part b only.

9 (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.



(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

(3)

percentage increase = ... %

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P 6 2 0 5 2 A 0 2 0 2 4

3. Question 9 · part b · 4 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

(1)

.....

.....

(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

(4)

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3. Question 9 · part b · 4 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

(c) There is evidence to show that glyphosate causes inflammation of the glomeruli and the kidney tubules.

Suggest how these symptoms will affect the normal functioning of the kidney and the waste product that it produces.

(4)

(Total for Question 9 = 12 marks)

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3. Question 9 · part b · 4 marks · Working and answer

Working and answer

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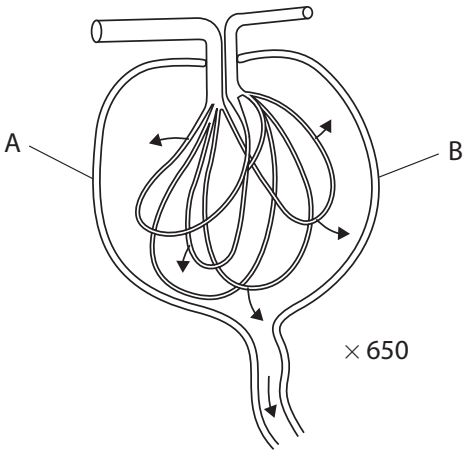
4. Question 1 · part a.i · 3 marks

January 2022 Paper 2

Further practice · Attempt part a.i only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

4. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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5. Question 7 · part a.i · 1 mark

June 2022 Paper 2R

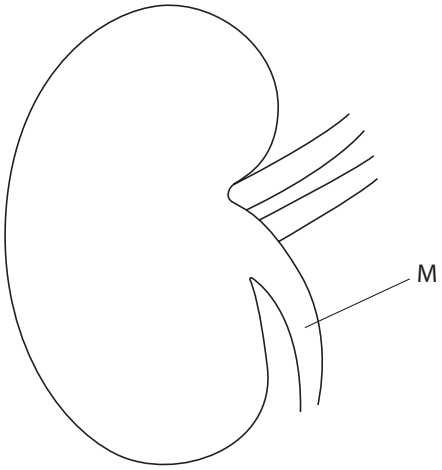
Further practice · Attempt part a.i only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

5. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

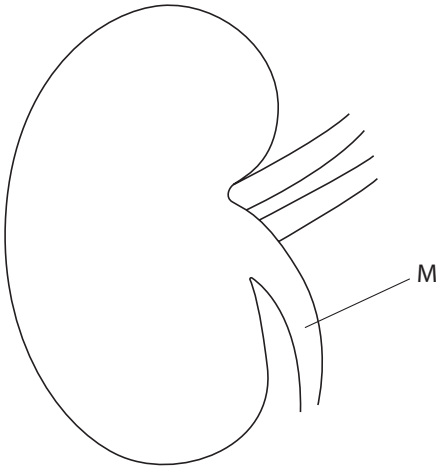
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6. Question 7 · part a.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

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6. Question 7 · part a.ii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part a.ii only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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P 7 0 9 5 0 A 0 2 2 2 4

6. Question 7 · part a.ii · 1 mark · Working and answer

Working and answer

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7. Question 7 · part a.ii · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

7 The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

(1)

.....

(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

(3)

	Structure		
Function	**glomerulus**	**proximal tubule**	**renal vein**
regulation of water level			
ultrafiltration			
reabsorption of glucose			
removal of blood from kidney			
supply of blood to kidney			
removal of urine from collecting duct			

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7. Question 7 · part a.ii · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.ii only.

(b) The table shows the percentage composition of two urine samples taken from one person.

Not all of the components are shown.

The first sample was taken at midday and the second sample was taken four hours later.

Component	Percentage composition	
	first sample	second sample
ammonia	0.05	0.08
uric acid	0.03	0.08
ions	1.34	1.46
urea	2.00	2.80
water	95.00	91.50

(i) Describe how urea is produced in the body.

(2)

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7. Question 7 · part a.ii · 3 marks · Working and answer

Working and answer

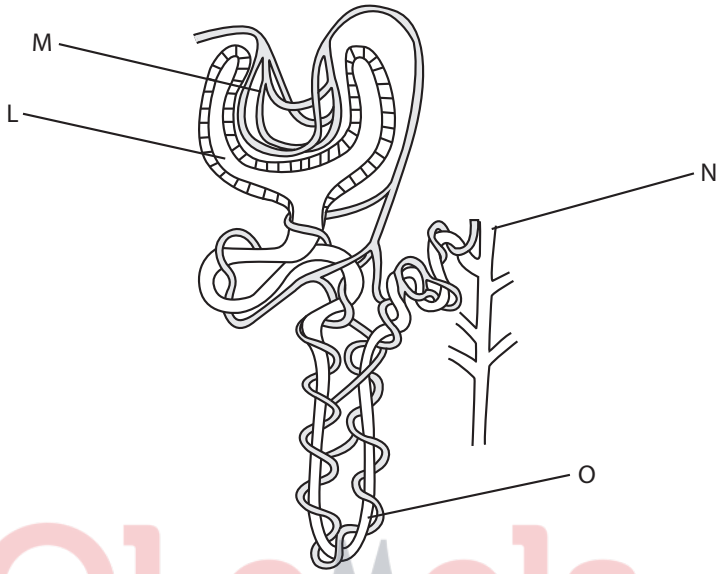
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8. Question 6 · part b.iii · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.iii only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O. **(4)**

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L. **(3)**

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8. Question 6 · part b.iii · 1 mark

November 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) Name **two** components of the blood that do not pass into L.

(2)

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

(1)

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

(5)

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(Total for Question 6 = 15 marks)

8. Question 6 · part b.iii · 1 mark · Working and answer

Working and answer

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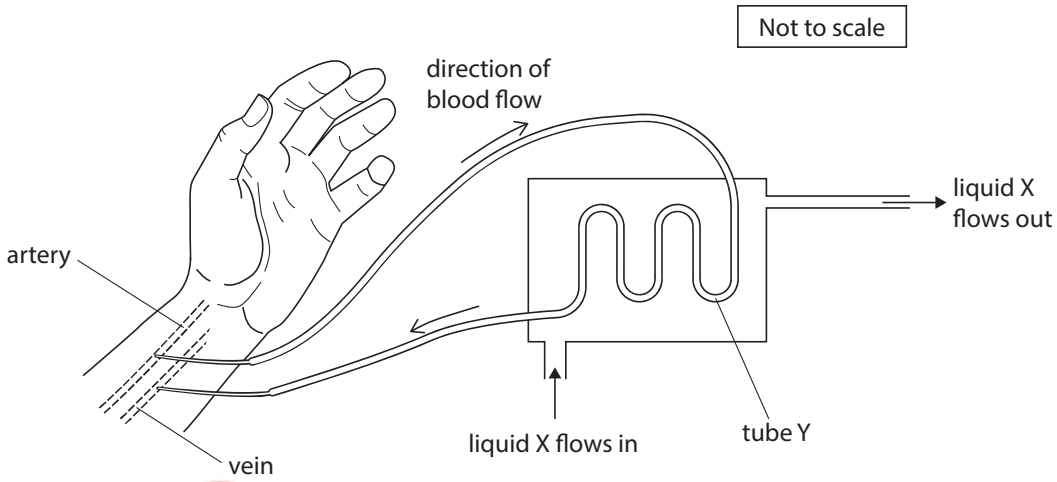
9. Question 5 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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P 7 4 6 2 8 A 0 1 2 2 0

9. Question 5 · part a.ii · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.ii only.

(iii) Explain why tube Y has a length of several metres in the dialysis machine.

(2)

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

(1)

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

(3)

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9. Question 5 · part a.ii · 1 mark · Working and answer

Working and answer

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Practice workbook · Explain homeostasis and negative feedback

Defining homeostasis, explaining negative feedback with temperature or glucose examples, and why a constant internal environment matters for enzymes.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-HOMEOSTASIS-NEGATIVE-FEEDBACK

1. Question 4 · part c · 4 marks

January 2022 Paper 1

Worked-example retry · Attempt part c only.

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4 Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

(3)

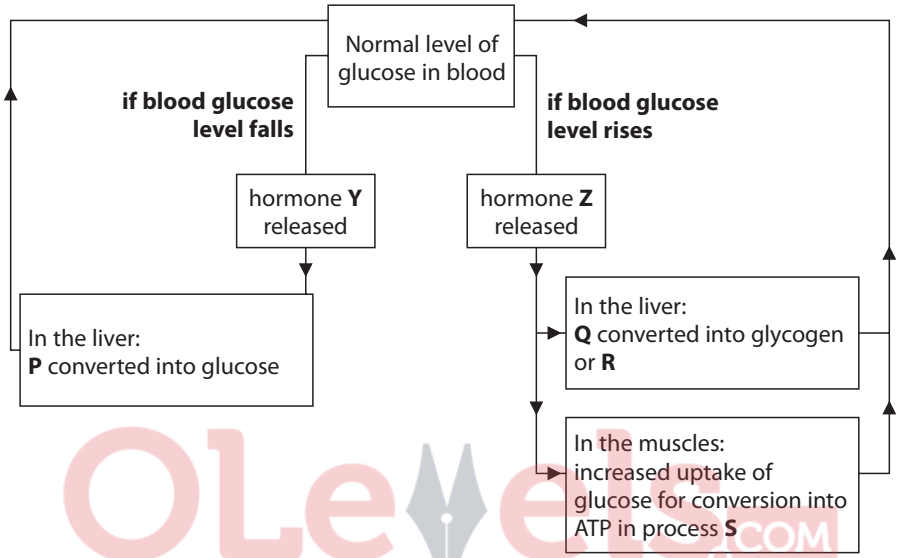
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1. Question 4 · part c · 4 marks

January 2022 Paper 1

Worked-example retry · Attempt part c only.

(b) The diagram shows the homeostatic control of blood glucose levels.



(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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1. Question 4 · part c · 4 marks

January 2022 Paper 1

Worked-example retry · Attempt part c only.

(iii) Identify P, Q, R and S.

(4)

P .....

Q .....

R .....

S .....

(c) The homeostatic process shown in the diagram is controlled by negative feedback.

Using the example shown in the diagram, explain what is meant by negative feedback.

(4)

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(Total for Question 4 = 13 marks)

1. Question 4 · part c · 4 marks · Working and answer

Working and answer

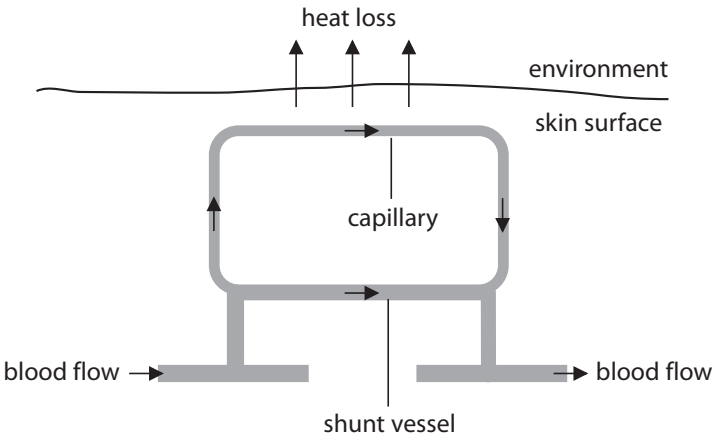
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2. Question 7 · part c · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

7 The diagram shows a section through skin on a cold day.



(a) Draw a diagram in the space to show how the section of skin shown in the diagram would be different on a hot day.

(2)

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2. Question 7 · part c · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

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(b) Explain why the section of skin shown in the diagram would be different on a hot day.

(4)

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(c) Explain how the control of body temperature is an example of negative feedback.

(2)

(Total for Question 7 = 8 marks)

2. Question 7 · part c · 2 marks · Working and answer

Working and answer

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3. Question 4 · part a · 3 marks

January 2022 Paper 1

Further practice · Attempt part a only.

4 Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

(3)

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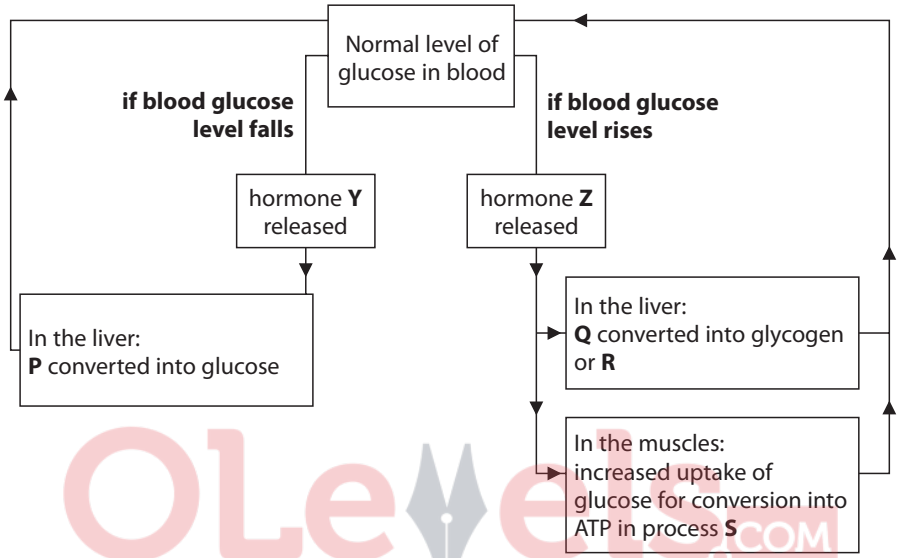
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3. Question 4 · part a · 3 marks

January 2022 Paper 1

Further practice · Attempt part a only.

(b) The diagram shows the homeostatic control of blood glucose levels.



(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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P 6 7 0 5 9 A 0 1 2 2 4

3. Question 4 · part a · 3 marks · Working and answer

Working and answer

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4. Question 3 · part d · 3 marks

January 2022 Paper 2

Further practice · Attempt part d only.

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3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

.....

2 .....

.....

(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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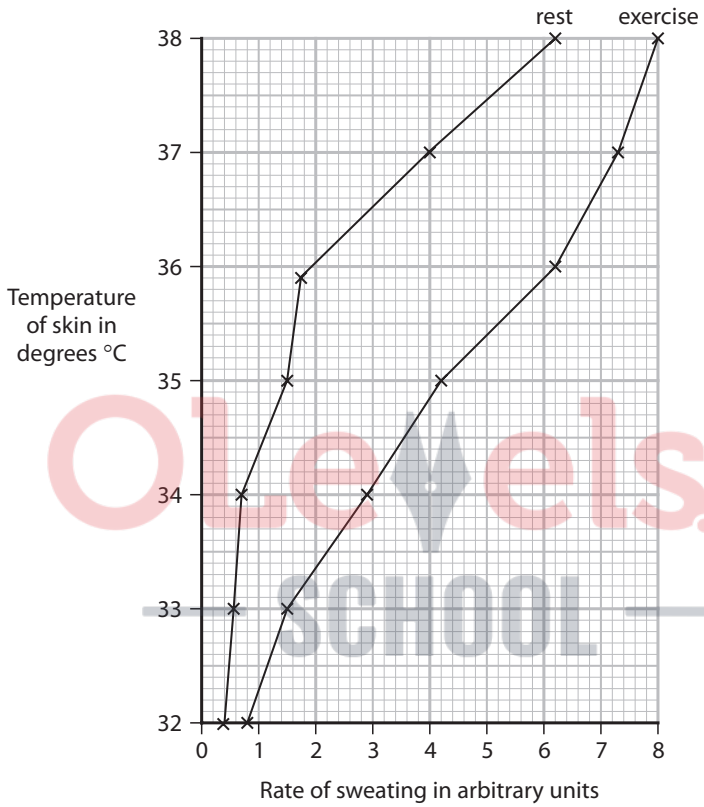
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4. Question 3 · part d · 3 marks

January 2022 Paper 2

Further practice · Attempt part d only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



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4. Question 3 · part d · 3 marks

January 2022 Paper 2

Further practice · Attempt part d only.

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(i) Describe how the skin temperature varies as the rate of sweating changes for the person at rest.

(2)

(ii) Suggest a method that could be used to measure the rate of sweating.

(2)

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(d) Explain the importance of maintaining a constant body temperature.

(3)

(Total for Question 3 = 13 marks)

4. Question 3 · part d · 3 marks · Working and answer

Working and answer

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5. Question 8 · part b.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

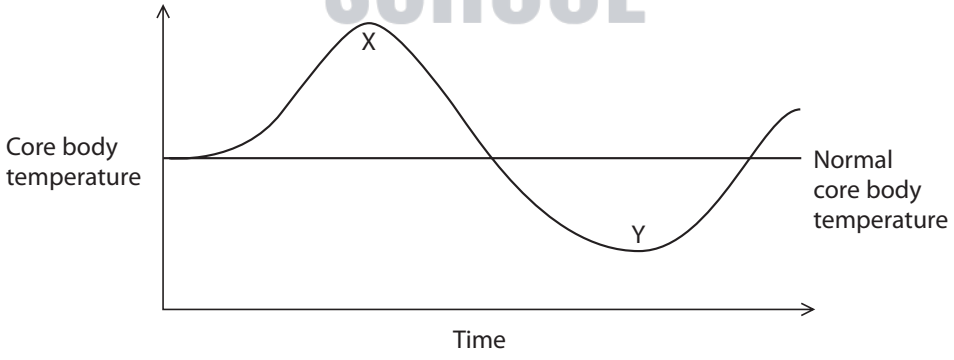
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

(3)

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.



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5. Question 8 · part b.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

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(i) Sweating can occur at point X on the graph.

Explain one other body process that occurs at point X to change body temperature.

(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

(iii) Explain why the control of body temperature is an example of negative feedback.

(2)

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5. Question 8 · part b.iii · 2 marks

June 2024 Paper 2R

Further practice · Attempt part b.iii only.

(c) Hypothermia is a condition where the core body temperature drops below 35 °C.

Explain why it is important to increase the core body temperature of people with hypothermia to within the normal range.

(5)

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(Total for Question 8 = 14 marks)

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5. Question 8 · part b.iii · 2 marks · Working and answer

Working and answer

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6. Question 3 · part a · 4 marks

June 2025 Paper 1

Further practice · Attempt part a only.

3 The control of body temperature is achieved by a process known as homeostasis.

(a) Explain what is meant by homeostasis.

(4)

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6. Question 3 · part a · 4 marks

June 2025 Paper 1

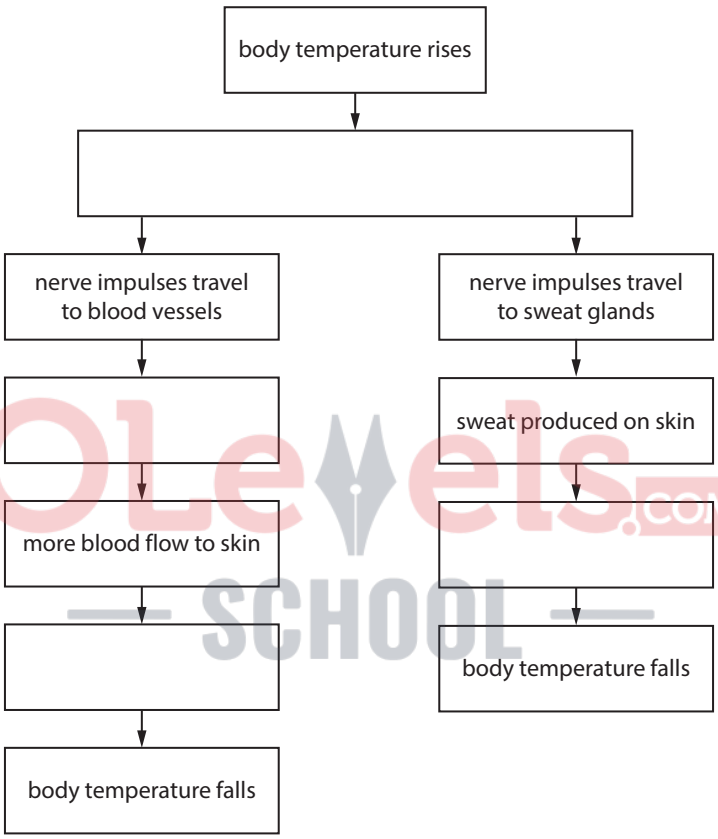
Further practice · Attempt part a only.

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(b) The diagram shows some of the reactions of the body to an increase in temperature.



Complete the diagram by putting in a suitable word or sentence in each of the blank boxes.

(6)

6. Question 3 · part a · 4 marks · Working and answer

Working and answer

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Practice workbook · Explain how insulin and glucagon control blood glucose

Pancreatic hormones, liver glycogen storage and release, and completing glucose control diagrams, including naming the hormones and glucose as the blood sugar.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BLOOD-GLUCOSE-INSULIN-GLUCAGON

1. Question 8 · part a · 3 marks

June 2024 Paper 2R

Worked-example retry · Attempt part a only.

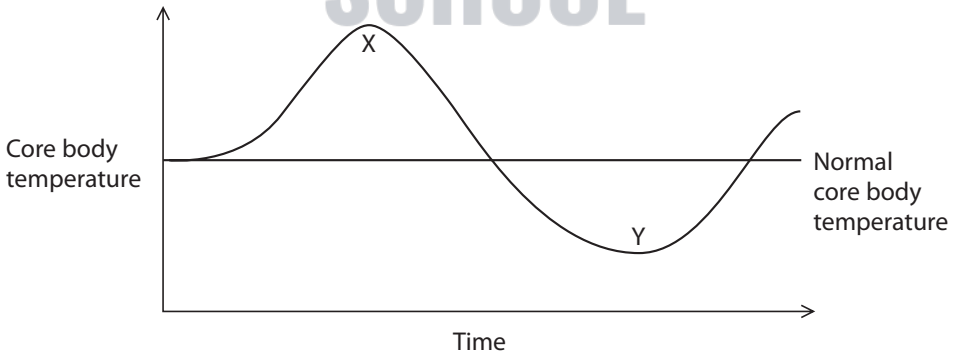
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

(3)

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.



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1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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2. Question 7 · part b.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

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2 .....

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(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

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2 .....

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2. Question 7 · part b.i · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) (i) Individuals who are overweight or obese have an increased risk of diabetes.

People with diabetes can have high blood glucose levels.

Explain how blood glucose levels are reduced in a person who does not have diabetes.

(3)

(ii) Suggest why one symptom of diabetes is frequent urination.

(2)

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(Total for Question 7 = 11 marks)

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2. Question 7 · part b.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part b.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.ii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Obesity and diabetes have been linked to a 20% increase in the number of leg and foot amputations.

Data from Diabetes UK shows that there were 26 378 lower limb amputations related to diabetes from the period 2014 to 2017. This is an increase from the period 2010 to 2013 when there were 22 092 amputations related to diabetes.

Minor lower limb amputations are below the ankle and major lower limb amputations are below the knee. Minor lower limb amputations increased by 26.5% and major lower limb amputations increased by 4.0%.

In 2019 in the United Kingdom 3.8 million people had been diagnosed with diabetes and 90% of these people had Type 2 diabetes.

(a) (i) Describe the causes of obesity.

(3)

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(ii) Give two health risks, other than diabetes, associated with obesity.

(2)

1 .....

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2 .....

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3. Question 4 · part b.ii · 1 mark

June 2021 Paper 2

Further practice · Attempt part b.ii only.

(b) Diabetes is caused as a result of insulin not being secreted or not working correctly.

(i) Which organ secretes insulin?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

(ii) Insulin stimulates the conversion of glucose into glycogen.

In which organ does this occur?

(1)

- ☒ **A** kidney
- ☒ **B** liver
- ☒ **C** pancreas
- ☒ **D** thyroid

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(c) (i) Calculate the percentage increase in the number of lower limb amputations between the period 2010 to 2013 and the period 2014 to 2017.

(3)

percentage increase = ... %

(ii) Calculate the number of people in 2019 who had been diagnosed with Type 2 diabetes in the United Kingdom.

(1)

number of people = ... million

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3. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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4. Question 4 · part b.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.i only.

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4 Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

(3)

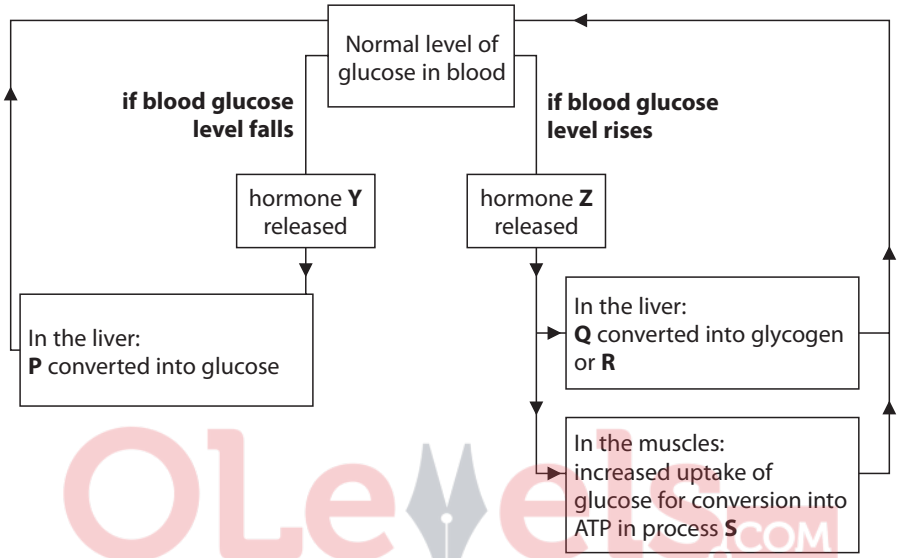
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4. Question 4 · part b.i · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.i only.

(b) The diagram shows the homeostatic control of blood glucose levels.



(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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P 6 7 0 5 9 A 0 1 2 2 4

4. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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5. Question 4 · part b.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.ii only.

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4 Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

(3)

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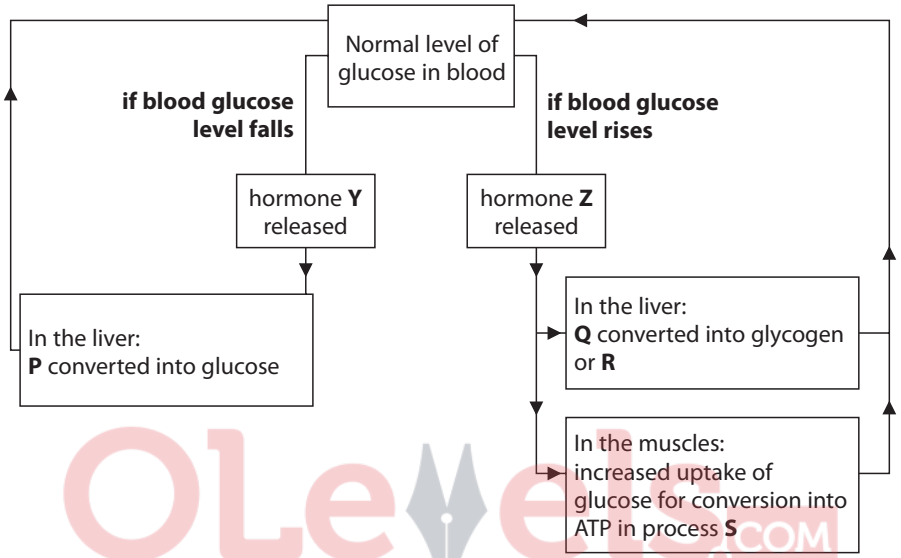
P 6 7 0 5 9 A 0 1 1 2 4

5. Question 4 · part b.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(b) The diagram shows the homeostatic control of blood glucose levels.



(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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P 6 7 0 5 9 A 0 1 2 2 4

5. Question 4 · part b.ii · 1 mark

January 2022 Paper 1

Further practice · Attempt part b.ii only.

(iii) Identify P, Q, R and S.

(4)

P .....

Q .....

R .....

S .....

(c) The homeostatic process shown in the diagram is controlled by negative feedback.

Using the example shown in the diagram, explain what is meant by negative feedback.

(4)

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(Total for Question 4 = 13 marks)

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5. Question 4 · part b.ii · 1 mark · Working and answer

Working and answer

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6. Question 4 · part b.iii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.iii only.

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4 Metabolic processes in the body are controlled by homeostasis.

(a) Explain what is meant by the term **homeostasis**.

(3)

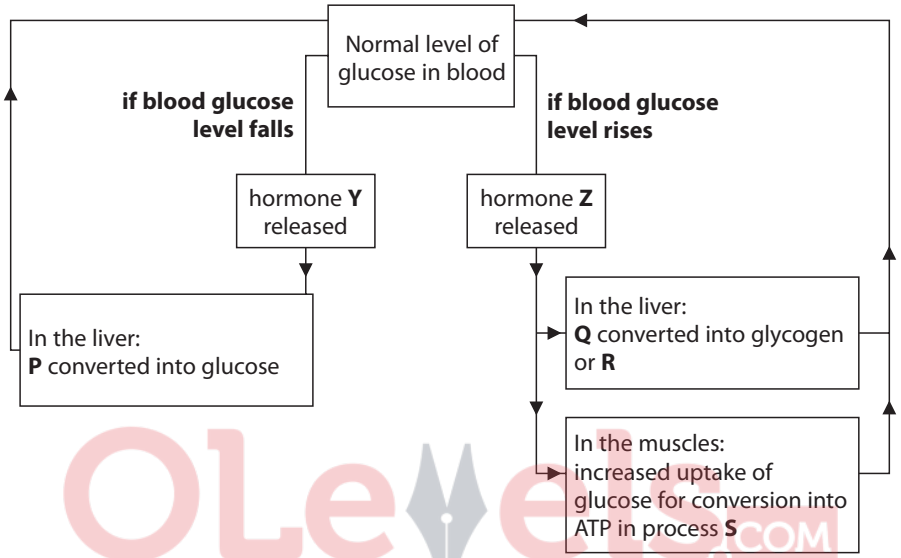
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6. Question 4 · part b.iii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.iii only.

(b) The diagram shows the homeostatic control of blood glucose levels.



(i) Which of these is the name of hormone Y?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

(ii) Which of these is the name of hormone Z?

(1)

- ☒ **A** ADH
- ☒ **B** glucagon
- ☒ **C** insulin
- ☒ **D** testosterone

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P 6 7 0 5 9 A 0 1 2 2 4

6. Question 4 · part b.iii · 4 marks

January 2022 Paper 1

Further practice · Attempt part b.iii only.

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(iii) Identify P, Q, R and S.

(4)

P .....

Q .....

R .....

S .....

(c) The homeostatic process shown in the diagram is controlled by negative feedback.

Using the example shown in the diagram, explain what is meant by negative feedback.

(4)

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(Total for Question 4 = 13 marks)

6. Question 4 · part b.iii · 4 marks · Working and answer

Working and answer

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7. Question 7 · part b · 1 mark

June 2022 Paper 1

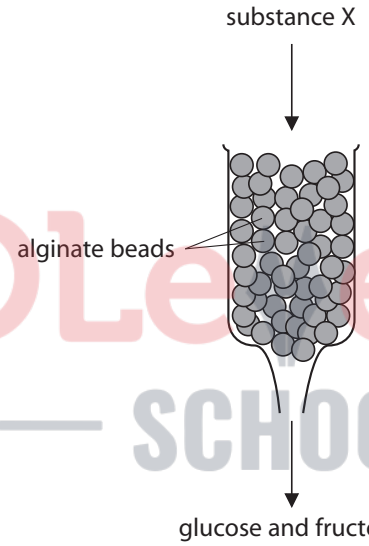
Further practice · Attempt part b only.

7 (a) Industrial processes can use immobilised enzymes.

(i) Give one reason why using immobilised enzymes can cost less money than using enzymes that are not immobilised.

(1)

(ii) The diagram shows the apparatus used to produce glucose and fructose using immobilised enzymes attached to alginate beads.



Name substance X.

(1)

(iii) Describe how to prepare alginate beads.

(3)

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P 7 0 9 7 3 A 0 2 0 2 4

7. Question 7 · part b · 1 mark

June 2022 Paper 1

Further practice · Attempt part b only.

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(b) Which process is involved in the conversion of glucose into glycogen in the human body?

(1)

- ☒ **A** glucagon is released from the pancreas and travels to the liver
- ☒ **B** insulin is released from the pancreas and travels to the liver
- ☒ **C** glucagon is released from the liver and travels to the pancreas
- ☒ **D** insulin is released from the liver and travels to the pancreas

(c) (i) The total population of the United Kingdom is 66.65 million.

Diabetes affects 6.0% of this population.

Calculate the number of people with diabetes in the United Kingdom.

(2)

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number of people = ... million

(ii) Surveys were carried out in different areas of the United Kingdom.

These surveys collected data on the number of people in each area with diabetes.

The information collected is shown in the box.

38	62	51	96	49	78	82	91	65

Determine the median number of people with diabetes.

(2)

median = ...

(Total for Question 7 = 10 marks)

7. Question 7 · part b · 1 mark · Working and answer

Working and answer

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8. Question 3 · part c.i · 1 mark

June 2023 Paper 2

Further practice · Attempt part c.i only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

.....

.....

.....

.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

8. Question 3 · part c.i · 1 mark

June 2023 Paper 2

Further practice · Attempt part c.i only.

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

(1)

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

(2)

(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

(3)

(Total for Question 3 = 13 marks)

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8. Question 3 · part c.i · 1 mark · Working and answer

Working and answer

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9. Question 4 · part a.ii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.ii only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

(2)

(ii) Name the hormone that stimulates the liver to produce glycogen.

(1)

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

(1)

(b) Consuming a diet high in carbohydrates can increase the BMI of a person.

The table gives information on the BMI categories of individuals.

BMI	Category
less than 18.5	underweight
18.5 to 24.9	healthy weight
25 to 29.9	overweight
30 to 34.9	obese
35 to 39.9	severely obese
more than 39.9	morbidly obese

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9. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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10. Question 4 · part a.iii · 1 mark

November 2024 Paper 1

Further practice · Attempt part a.iii only.

4 Carbohydrates are large molecules found in many body cells.

(a) Glycogen is a carbohydrate.

(i) Describe how the liver produces glycogen.

(2)

(ii) Name the hormone that stimulates the liver to produce glycogen.

(1)

(iii) Name a hormone that stimulates the liver to increase blood sugar levels.

(1)

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The table gives information on the BMI categories of individuals.

BMI	Category
less than 18.5	underweight
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10. Question 4 · part a.iii · 1 mark · Working and answer

Working and answer

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11. Question 3 · part c.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part c.ii only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

.....

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.....

.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

11. Question 3 · part c.ii · 2 marks

June 2023 Paper 2

Further practice · Attempt part c.ii only.

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

(1)

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

(2)

(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

(3)

(Total for Question 3 = 13 marks)

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11. Question 3 · part c.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how skin blood flow and sweating control body temperature

Vasodilation, vasoconstriction, shivering and sweating in hot, cold or exercise conditions, including drawn skin changes and differences in sweat volume between situations or people.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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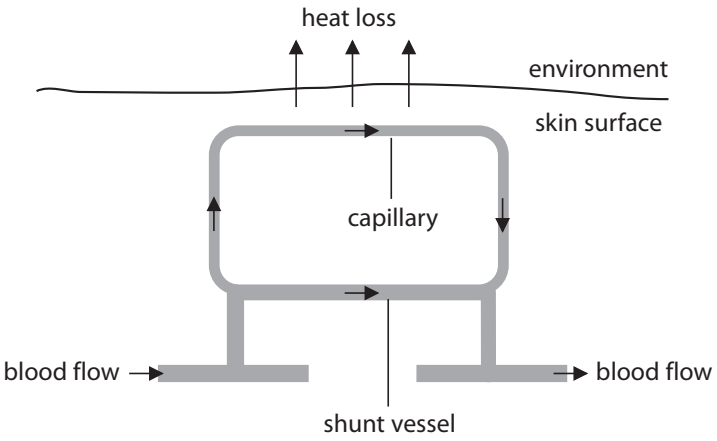
Type ID: HBI-SKIN-THERMOREGULATION-RESPONSES

1. Question 7 · part b · 4 marks

November 2020 Paper 1R

Worked-example retry · Attempt part b only.

7 The diagram shows a section through skin on a cold day.



(a) Draw a diagram in the space to show how the section of skin shown in the diagram would be different on a hot day.

(2)

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1. Question 7 · part b · 4 marks

November 2020 Paper 1R

Worked-example retry · Attempt part b only.

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(b) Explain why the section of skin shown in the diagram would be different on a hot day.

(4)

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(c) Explain how the control of body temperature is an example of negative feedback.

(2)

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(Total for Question 7 = 8 marks)

1. Question 7 · part b · 4 marks · Working and answer

Working and answer

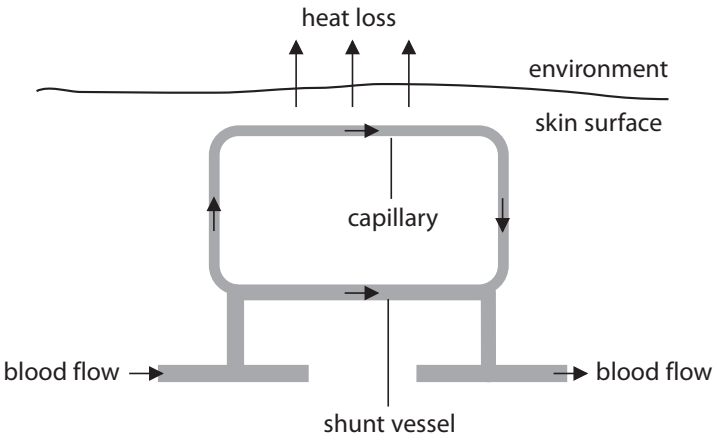
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2. Question 7 · part a · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a only.

7 The diagram shows a section through skin on a cold day.



(a) Draw a diagram in the space to show how the section of skin shown in the diagram would be different on a hot day.

(2)

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2. Question 7 · part a · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a only.

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(b) Explain why the section of skin shown in the diagram would be different on a hot day.

(4)

(c) Explain how the control of body temperature is an example of negative feedback.

(2)

(Total for Question 7 = 8 marks)

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2. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part b · 4 marks

January 2022 Paper 2

Further practice · Attempt part b only.

3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

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2 .....

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(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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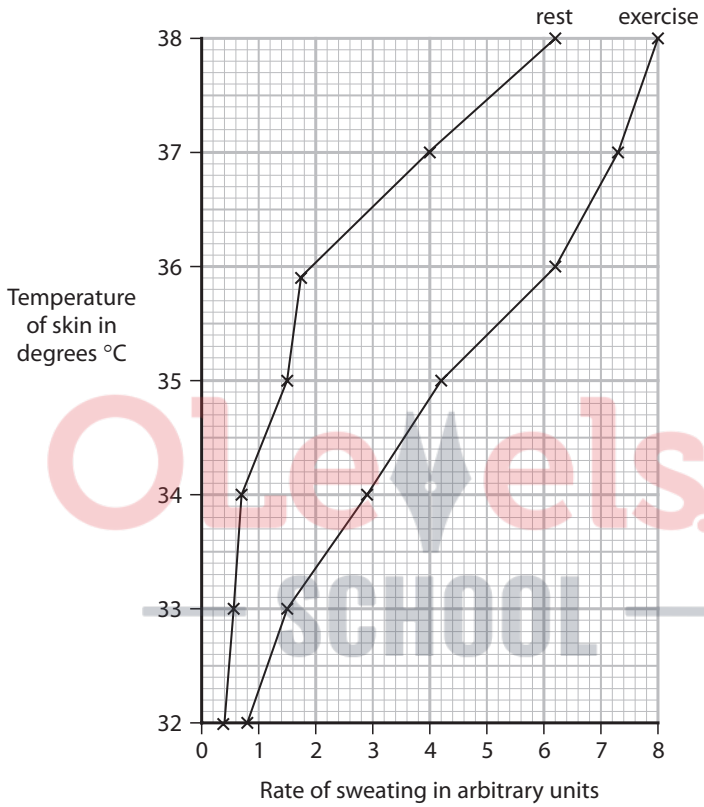
OLevels.com SCHOOL

3. Question 3 · part b · 4 marks

January 2022 Paper 2

Further practice · Attempt part b only.

(c) The graph shows how the temperature of the skin varies as the rate of sweating changes in a person at rest and during exercise.



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3. Question 3 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 5 · part b · 3 marks

June 2022 Paper 1

Further practice · Attempt part b only.

5 (a) There is a difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Design an investigation to determine the difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Include experimental details in your answer and write in full sentences.

(4)

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4. Question 5 · part b · 3 marks

June 2022 Paper 1

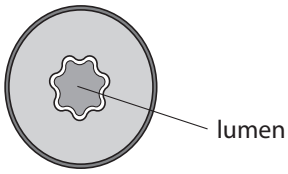
Further practice · Attempt part b only.

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(b) The diagram shows a cross-section of a blood vessel in the skin on a cold day.



Explain the changes in this blood vessel on a hot day.

(3)

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(c) Describe how ADH regulates the volume of water in the body on a hot day.

(4)

(Total for Question 5 = 11 marks)

4. Question 5 · part b · 3 marks · Working and answer

Working and answer

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5. Question 3 · part b.ii · 4 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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5. Question 3 · part b.ii · 4 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

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(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

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(Total for Question 3 = 15 marks)

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5. Question 3 · part b.ii · 4 marks · Working and answer

Working and answer

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6. Question 8 · part b.i · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

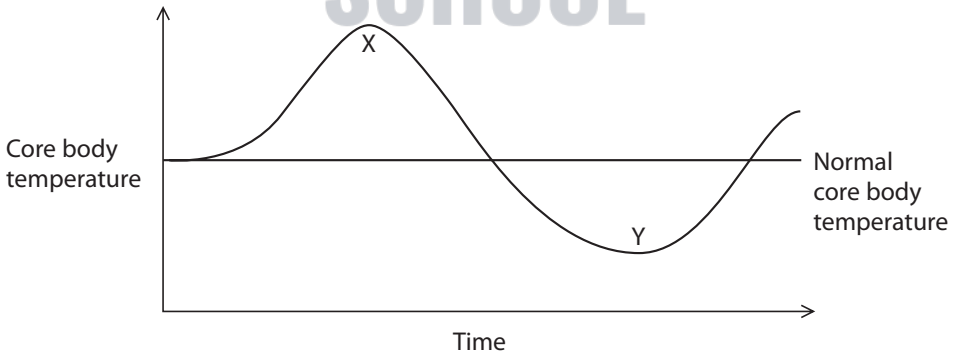
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

(3)

(b) The normal core body temperature of a human is 37 °C.

The graph shows how body temperature changes during one hour.



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P 7 4 4 9 5 A 0 2 4 2 8

6. Question 8 · part b.i · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.i only.

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(i) Sweating can occur at point X on the graph.

Explain one other body process that occurs at point X to change body temperature.

(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

(iii) Explain why the control of body temperature is an example of negative feedback.

(2)

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6. Question 8 · part b.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part b.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

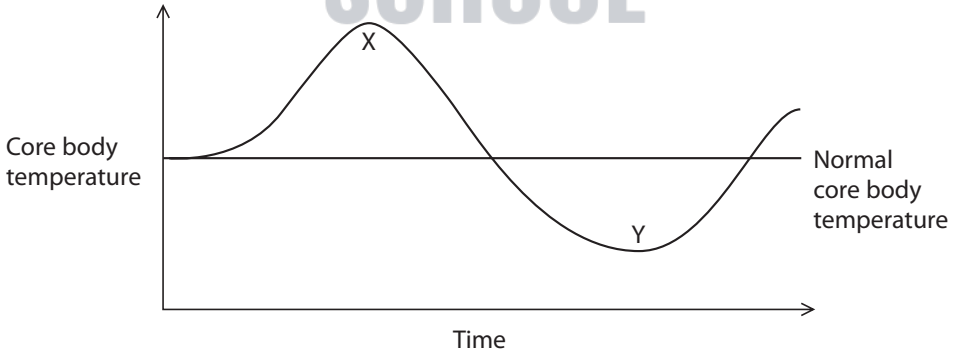
8 (a) Insulin is a hormone which helps to regulate the concentration of glucose in the blood.

Explain the role of insulin in regulating the concentration of glucose in the blood.

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The graph shows how body temperature changes during one hour.



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P 7 4 4 9 5 A 0 2 4 2 8

7. Question 8 · part b.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

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(i) Sweating can occur at point X on the graph.

Explain one other body process that occurs at point X to change body temperature.

(3)

(ii) Name one body process that occurs at point Y on the graph to change body temperature.

(1)

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(2)

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7. Question 8 · part b.ii · 1 mark · Working and answer

Working and answer

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8. Question 3 · part b · 6 marks

June 2025 Paper 1

Further practice · Attempt part b only.

3 The control of body temperature is achieved by a process known as homeostasis.

(a) Explain what is meant by homeostasis.

(4)

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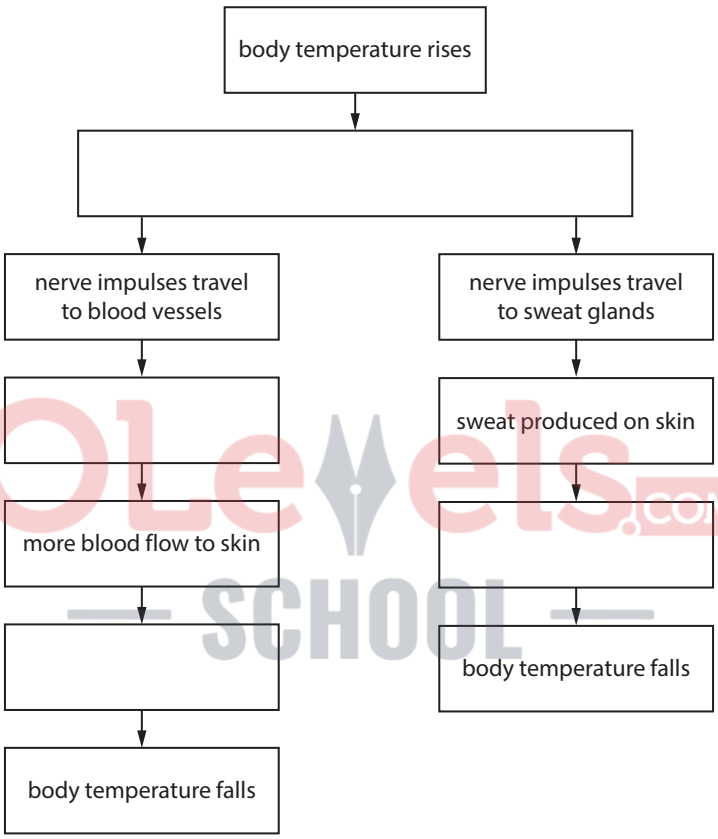
OLevels.COM SCHOOL

8. Question 3 · part b · 6 marks

June 2025 Paper 1

Further practice · Attempt part b only.

(b) The diagram shows some of the reactions of the body to an increase in temperature.



Complete the diagram by putting in a suitable word or sentence in each of the blank boxes.

(6)

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8. Question 3 · part b · 6 marks

June 2025 Paper 1

Further practice · Attempt part b only.

(c) Explain why it is important that body temperature should not become too high.

(4)

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(Total for Question 3 = 14 marks)

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8. Question 3 · part b · 6 marks · Working and answer

Working and answer

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9. Question 5 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

5 A student investigates the effect of temperature on the volume of sweat produced.

This is the student's method.

- sit a person in a room at 20 °C
- measure the volume of sweat the person produces
- repeat with the room at different temperatures
- repeat again when the person is active

The table shows the student's results.

Temperature of room in °C	Volume of sweat produced in arbitrary units	
	Sitting	Active
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

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9. Question 5 · part c · 3 marks

June 2019 Paper 1R

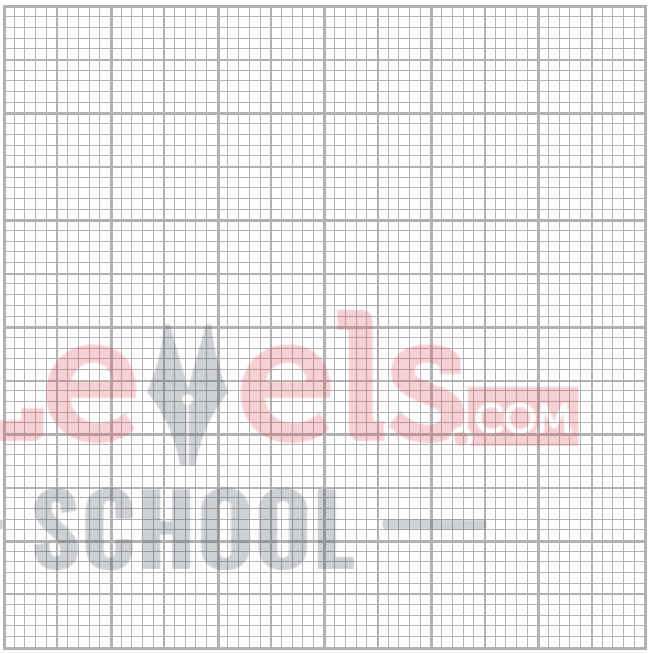
Further practice · Attempt part c only.

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(a) Plot the student's results on the grid.

Join the points with straight lines.

(5)



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(b) Use your graph to determine the difference in the volume of sweat produced at 27 °C when sitting and when active.

(2)

difference = ... arbitrary units

P 6 0 1 8 6 A 0 1 3 2 8

9. Question 5 · part c · 3 marks

June 2019 Paper 1R

Further practice · Attempt part c only.

(c) Explain the difference between the volume of sweat produced when sitting and when active.

(3)

(Total for Question 5 = 10 marks)

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Working and answer

10. Question 3 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Sweating is normal when we are hot. It is part of our natural cooling mechanism. If you sweat visibly when you do not need to cool down, it can be a problem.

The sweating process is controlled by the brain, which sends signals down the nerves to the sweat glands.

Sweating is crucial to survival. We sweat to control body temperature and the sweat glands in our armpits can produce several litres of sweat in 24 hours. How much we sweat varies hugely from individual to individual and even from day to day.

Hyperhidrosis occurs when some areas of the body start sweating excessively. This affects about 1% of the population. There are two main types. The most common type is primary focal hyperhidrosis, which affects many parts of the body. There is no known cause, but it usually begins in childhood and often runs in families. Another type is secondary hyperhidrosis, which often doesn't begin until after the teenage years, and usually has an underlying medical cause. These causes can include diabetes, menopause or chronic infection. It can also be a result of eating certain foods. Stress can play a major role too.

(a) Which part of the brain controls sweating (line 4)?

(1)

- ☒ **A** cerebellum
- ☒ **B** cerebral hemispheres
- ☒ **C** hypothalamus
- ☒ **D** pituitary gland

(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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10. Question 3 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

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(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

10. Question 3 · part c · 3 marks · Working and answer

Working and answer

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11. Question 3 · part f · 5 marks

November 2021 Paper 2

Further practice · Attempt part f only.

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(b) Which type of nerve sends signals to the sweat glands (lines 4 and 5)?

(1)

- ☒ **A** connector
- ☒ **B** motor
- ☒ **C** relay
- ☒ **D** sensory

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11. Question 3 · part f · 5 marks

November 2021 Paper 2

Further practice · Attempt part f only.

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(c) Describe how sweating controls body temperature (lines 6 and 7).

(3)

(d) Explain the evidence in the passage that indicates primary focal hyperhidrosis is a genetic condition.

(3)

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(e) Secondary hyperhidrosis can be caused by diabetes.

Give one risk factor for diabetes (line 16).

(1)

11. Question 3 · part f · 5 marks

November 2021 Paper 2

Further practice · Attempt part f only.

(f) The passage states that how much we sweat varies from individual to individual and from day to day.

Explain why this is the case in people without hyperhidrosis (lines 8 and 9).

(5)

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(Total for Question 3 = 14 marks)

11. Question 3 · part f · 5 marks · Working and answer

Working and answer

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12. Question 5 · part b.ii · 5 marks

June 2023 Paper 1R

Further practice · Attempt part b.ii only.

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5 A student uses this method to investigate the effect of exercise on the production of sweat.

- place a person in a room which has a temperature of 20 °C
- ask the person to remain inactive for 5 minutes
- measure the amount of sweat produced by a 25 cm^2 area of skin
- ask the person to exercise and measure the amount of sweat produced
- ask the person to rest for 10 minutes

Repeat the method at room temperatures of 22 °C, 24 °C, 26 °C, 28 °C and 30 °C

(a) (i) Explain why the person rests for 10 minutes before the method is repeated.

(2)

.....

.....

.....

.....

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(ii) State three variables that should be controlled.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

12. Question 5 · part b.ii · 5 marks

June 2023 Paper 1R

Further practice · Attempt part b.ii only.

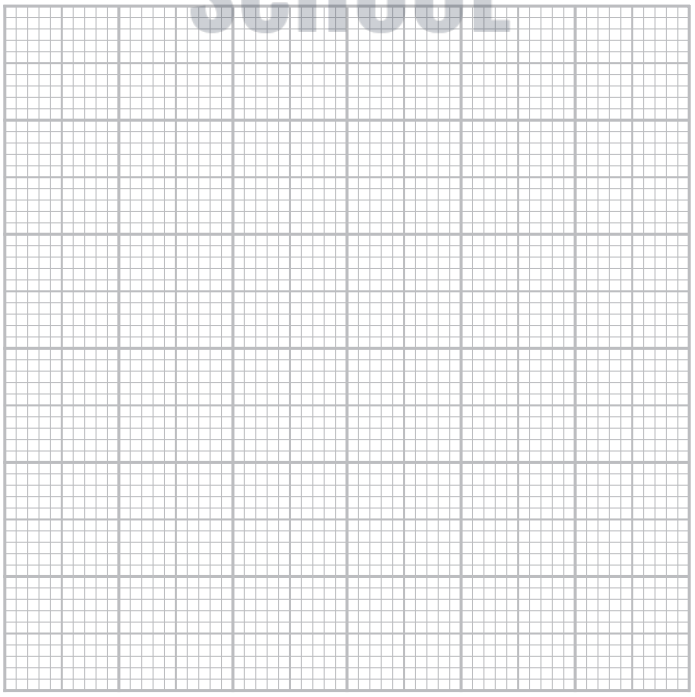
(b) The table shows the student's results.

Room temperature in °C	Amount of sweat produced by 25 cm^2 area of skin in arbitrary units	
	While inactive	During exercise
20	1	2
22	2	6
24	2	8
26	2	11
28	3	14
30	4	22

(i) Plot line graphs of the student's results.

Join the points with straight lines.

(6)



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P 7 2 5 8 6 A 0 1 2 2 0

12. Question 5 · part b.ii · 5 marks

June 2023 Paper 1R

Further practice · Attempt part b.ii only.

(ii) Explain the differences between the two sets of results.

(5)

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(Total for Question 5 = 16 marks)

12. Question 5 · part b.ii · 5 marks · Working and answer

Working and answer

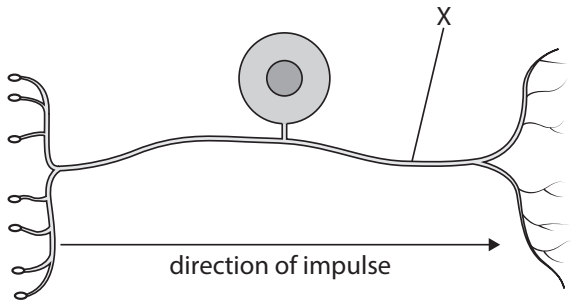
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13. Question 1 · part b.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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P 7 4 6 2 6 R A 0 2 2 4

13. Question 1 · part b.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

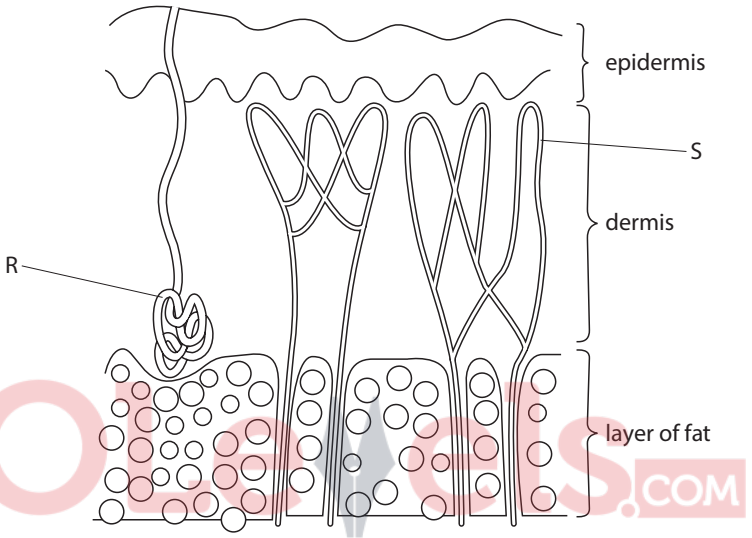
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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

.....

.....

.....

.....

13. Question 1 · part b.ii · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.ii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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13. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · Explain the liver's roles in deamination and breaking down drugs

Deamination of surplus amino acids to urea, why kidney patients limit protein, the liver's large blood supply, and why liver damage lets alcohol or medicines build up.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-LIVER-DEAMINATION-AND-DETOXIFICATION

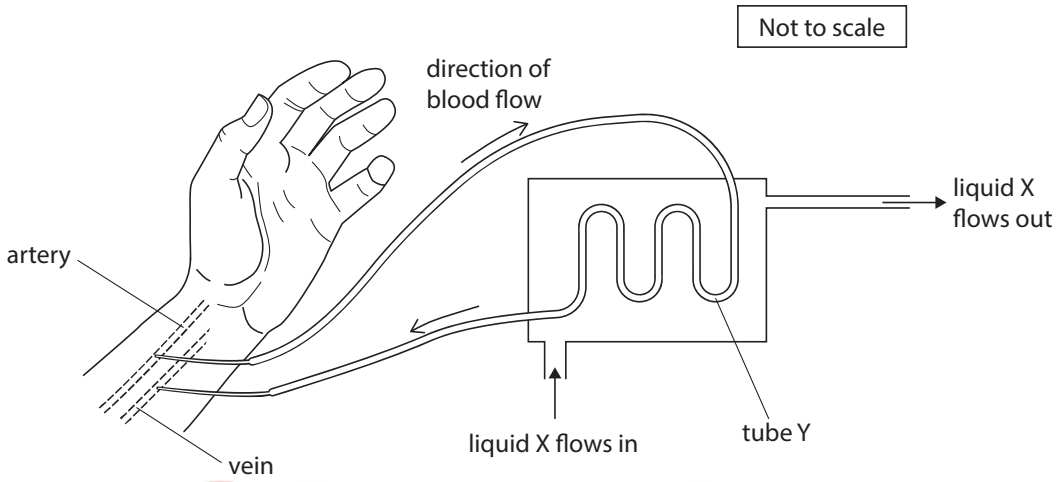
1. Question 5 · part c · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part c only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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1. Question 5 · part c · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part c only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

(2)

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

(1)

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

(3)

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1. Question 5 · part c · 4 marks

June 2025 Paper 1R

Worked-example retry · Attempt part c only.

(c) Explain why people who have to use kidney dialysis machines are often told to eat less protein in their diet.

(4)

(Total for Question 5 = 15 marks)

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Working and answer

2. Question 8 · part a.i · 3 marks

June 2019 Paper 2R

Further practice · Attempt part a.i only.

8 (a) Alkaptonuria is a condition caused by a gene mutation that produces a recessive allele.

People who have this condition are unable to fully break down two amino acids found in proteins. Instead, an intermediate compound is formed which gives a dark colour to the urine.

This intermediate compound causes damage to cartilage and heart valves.

(i) Explain how amino acids are normally metabolised.

(3)

(ii) A person with alkaptonuria produces a dark coloured urine.

Describe what other symptoms they may develop.

(3)

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2. Question 8 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 9 · part a.i · 3 marks

January 2020 Paper 2

Further practice · Attempt part a.i only.

9 (a) Phenylketonuria (PKU) is a serious genetic disorder.

People with PKU are unable to break down the amino acid, phenylalanine.

Phenylalanine builds up in the blood and in the brain causing brain damage.

(i) Describe how levels of phenylalanine are reduced in the body of a healthy person.

(3)

(ii) Explain why people with PKU are advised to follow a low-protein diet.

(3)

(iii) Explain the impact of a low-protein diet on protein synthesis in the cytoplasm of body cells.

(3)

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3. Question 9 · part a.i · 3 marks · Working and answer

Working and answer

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4. Question 7 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

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7 The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

(1)

.....

(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

(3)

	Structure		
Function	**glomerulus**	**proximal tubule**	**renal vein**
regulation of water level			
ultrafiltration			
reabsorption of glucose			
removal of blood from kidney			
supply of blood to kidney			
removal of urine from collecting duct			

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4. Question 7 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(b) The table shows the percentage composition of two urine samples taken from one person.

Not all of the components are shown.

The first sample was taken at midday and the second sample was taken four hours later.

Component	Percentage composition	
	first sample	second sample
ammonia	0.05	0.08
uric acid	0.03	0.08
ions	1.34	1.46
urea	2.00	2.80
water	95.00	91.50

(i) Describe how urea is produced in the body.

(2)

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4. Question 7 · part b.i · 2 marks

January 2023 Paper 2

Further practice · Attempt part b.i only.

(ii) The ADH response changes the percentage composition of urine.

Explain the ADH response that occurred in the person's body between the first and the second sample being collected.

(5)

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(Total for Question 7 = 11 marks)

4. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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5. Question 7 · part b · 3 marks

June 2023 Paper 1

Further practice · Attempt part b only.

7 A method that shows whether a person's kidneys are working correctly is to measure the volume of filtrate produced by the kidneys in one minute. This is known as the glomerular filtration rate (GFR).

A person with healthy kidneys has a GFR of over 100 cm^3 per minute.

The GFR of a person whose kidneys were not working correctly was measured once a year for nine years.

The table shows the results.

Year	2011	2012	2013	2014	2015	2016	2017	2018	2019
GFR in cm^3 per minute	80	75	79	79	67	60	50	45	35

(a) (i) Plot a line graph of the data on the grid.

(4)



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5. Question 7 · part b · 3 marks

June 2023 Paper 1

Further practice · Attempt part b only.

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(ii) Explain why it is not possible to determine from the data when the person's kidneys stopped working correctly.

(2)

(iii) A person needs dialysis if the GFR falls to 15 cm^3 per minute.

Use the data to estimate when the person will probably need to start dialysis.

(2)

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(iv) Discuss the accuracy of this estimate.

(3)

P 7 2 5 8 5 A 0 1 9 2 0

5. Question 7 · part b · 3 marks

June 2023 Paper 1

Further practice · Attempt part b only.

(b) During the early stages when a person's kidneys are not working correctly, they are advised to control the amount of protein in their diet.

Explain why the protein in their diet should be controlled.

(3)

(Total for Question 7 = 14 marks)

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5. Question 7 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

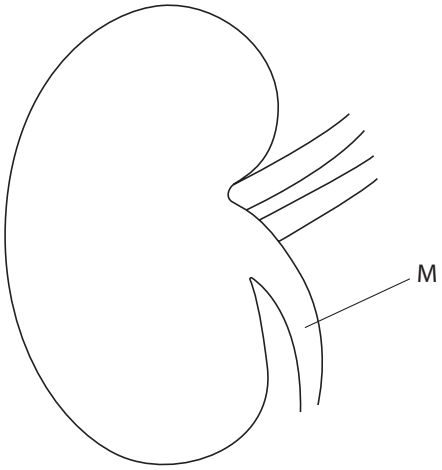
6. Question 7 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

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7 The diagram shows a kidney.



(a) (i) Which of these is the name of structure M?

(1)

- ☒ **A** renal artery
- ☒ **B** renal vein
- ☒ **C** ureter
- ☒ **D** urethra

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(ii) Which of these structures is connected to structure M?

(1)

- ☒ **A** bladder
- ☒ **B** prostate gland
- ☒ **C** seminal vesicle
- ☒ **D** urethra

6. Question 7 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

(b) State three differences between the blood entering the kidney and the blood leaving the kidney.

(3)

1 .....

.....

2 .....

.....

3 .....

.....

(c) The table compares the nitrogen content of urine during 24 hours for a person on a high protein diet with a person on a low protein diet.

Substance excreted during 24 hours	High protein diet Mass in g	Low protein diet Mass in g
total nitrogen compounds	16.8	3.6
urea	14.7	2.2
uric acid	0.18	0.09
ammonia	0.49	0.04
creatinine	0.58	0.60

(i) Calculate the percentage difference in the mass of nitrogen compounds excreted during 24 hours for the person on a high protein diet compared with the low protein diet.

(3)

percentage difference = ... %

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6. Question 7 · part c.ii · 2 marks

June 2022 Paper 2R

Further practice · Attempt part c.ii only.

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(ii) Which is the main nitrogen containing excretory product produced from the breakdown of protein in the diet?

Use the information in the table to explain your answer.

(2)

(iii) The breakdown of muscle tissue continues at a constant rate and is not affected by diet.

Discuss whether the information in the table supports this statement.

(4)

(Total for Question 7 = 14 marks)

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6. Question 7 · part c.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part b · 4 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

8 The table shows the rate of blood flow through different organs when a person is at rest. The table also shows the masses of these organs.

Organ	Mass in kg	Rate of blood flow in cm^3 per minute	
		through the whole organ	per 100g of the organ
kidneys	0.3	1200	
skeletal muscles	25.0		3
liver	1.5	1500	100

(a) Calculate the missing values and put them in the table.

(4)

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(b) Explain why the liver has a high rate of blood flow.

(4)

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7. Question 8 · part b · 4 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

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(c) Explain any changes that would occur in the blood flow through these organs during vigorous exercise.

(4)

(Total for Question 8 = 12 marks)

OLevels.com SCHOOL

Working and answer

8. Question 4 · part b.i · 4 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

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8. Question 4 · part b.i · 4 marks

November 2023 Paper 2

Further practice · Attempt part b.i only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

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8. Question 4 · part b.i · 4 marks · Working and answer

Working and answer

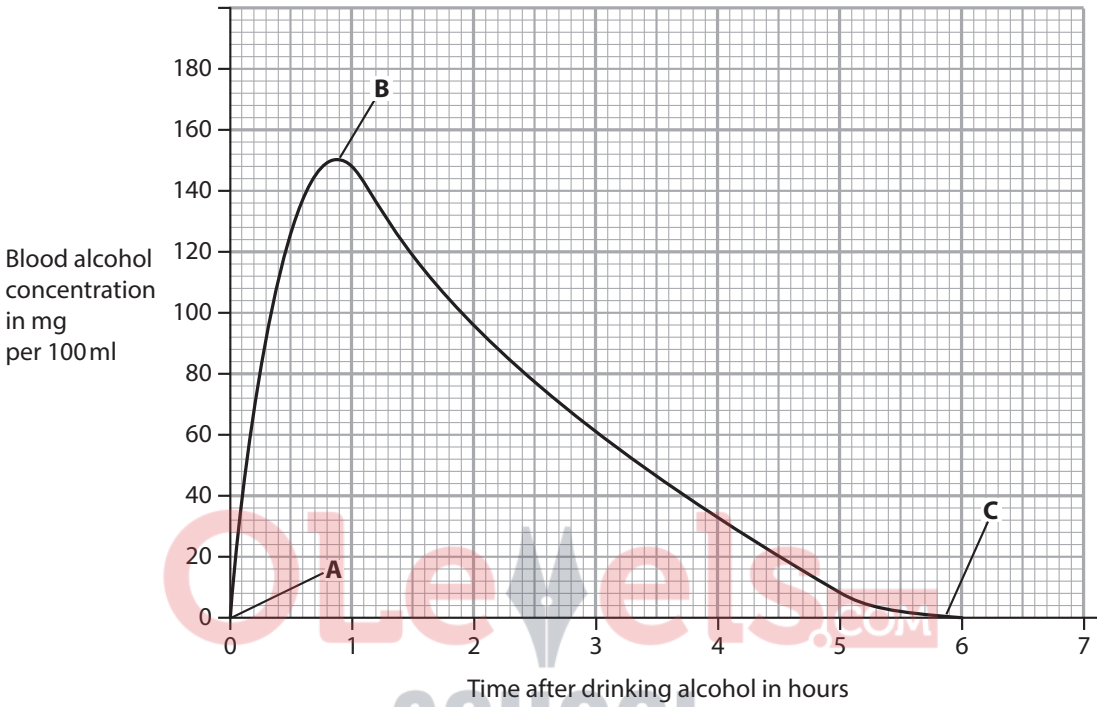
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9. Question 6 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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9. Question 6 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

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(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

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9. Question 6 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 5 · part e · 2 marks

June 2025 Paper 2

Further practice · Attempt part e only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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10. Question 5 · part e · 2 marks

June 2025 Paper 2

Further practice · Attempt part e only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

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(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

10. Question 5 · part e · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain ultrafiltration and selective reabsorption in the nephron

High pressure in the glomerulus forcing small molecules into the capsule, what stays in the blood, glucose reabsorption, and why diabetes or damage puts protein or glucose into urine.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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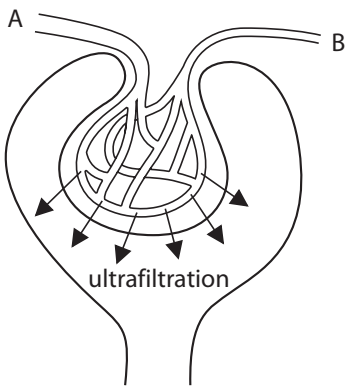
Type ID: HBI-KIDNEY-FILTRATION-REABSORPTION

1. Question 6 · part a.iii · 4 marks

June 2021 Paper 1

Worked-example retry · Attempt part a.iii only.

6 (a) The diagram shows a Bowman's capsule and its associated blood supply.



(i) Describe what is meant by the term **ultrafiltration**.

(2)

(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

(3)

1 .....

2 .....

3 .....

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1. Question 6 · part a.iii · 4 marks

June 2021 Paper 1

Worked-example retry · Attempt part a.iii only.

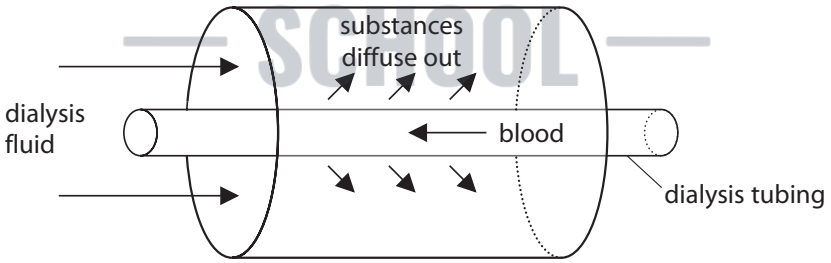
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(iii) Explain how blood vessel A and blood vessel B help the process of ultrafiltration.

(4)

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(b) The diagram shows part of a kidney dialysis machine.



The dialysis fluid contains glucose at the same concentration as in the blood flowing through the dialysis tubing.

Suggest the effect on a person undergoing dialysis if the concentration of glucose was below that of the blood flowing through the dialysis tubing.

(3)

(Total for Question 6 = 12 marks)

1. Question 6 · part a.iii · 4 marks · Working and answer

Working and answer

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2. Question 4 · part a.i · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a.i only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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2. Question 4 · part a.i · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 4 · part a.ii · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a.ii only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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3. Question 4 · part a.ii · 1 mark

June 2019 Paper 1R

Further practice · Attempt part a.ii only.

(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

percentage = ... %

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3. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

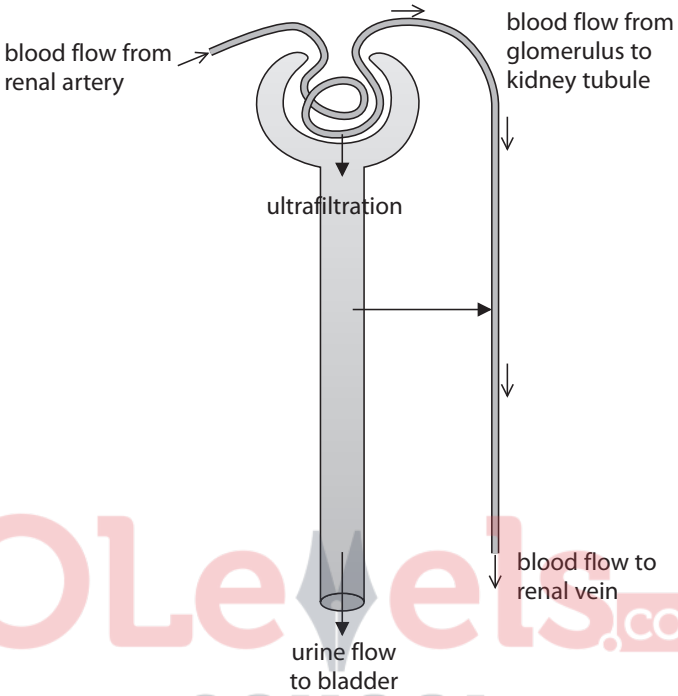
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4. Question 8 · part a.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

8 (a) The diagram shows part of a nephron in the human excretory system.



(Source: adapted from © Alila Medical Images/Alamy)

(i) Explain how ultrafiltration occurs in the Bowman's capsule.

(2)

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4. Question 8 · part a.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part a.i only.

(ii) Explain the differences in the composition of the blood flowing through the renal artery and through the renal vein.

(4)

(b) The quantity of salt in the diet affects the concentration of urine produced.

An investigation is set up to test this statement.

This is the method used.

- ten people are each given a different food
- each food contains a different quantity of salt
- the volume of urine produced by each person is recorded each hour for six hours after eating the food

(i) State the independent variable in this investigation.

(1)

(ii) Explain why this investigation is unlikely to produce valid results.

(2)

(iii) State one change to the method that would make the results more valid.

(1)

(Total for Question 8 = 10 marks)

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4. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

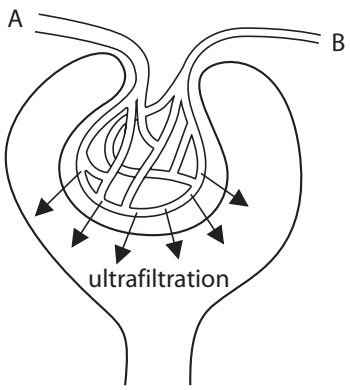
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5. Question 6 · part a.i · 2 marks

June 2021 Paper 1

Further practice · Attempt part a.i only.

6 (a) The diagram shows a Bowman's capsule and its associated blood supply.



(i) Describe what is meant by the term **ultrafiltration**.

(2)

.....

.....

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(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

(3)

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2 .....

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3 .....

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5. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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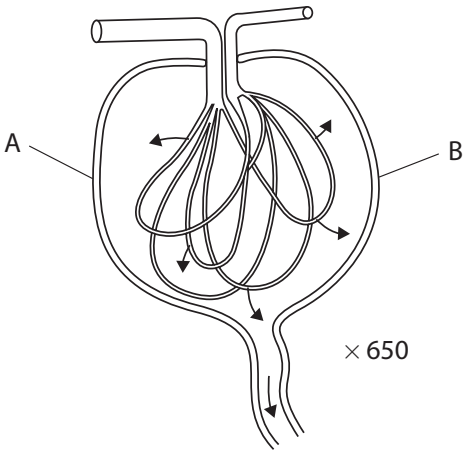
6. Question 1 · part a.iii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.iii only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

6. Question 1 · part a.iii · 2 marks

January 2022 Paper 2

Further practice · Attempt part a.iii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
0.5	70	50
1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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6. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

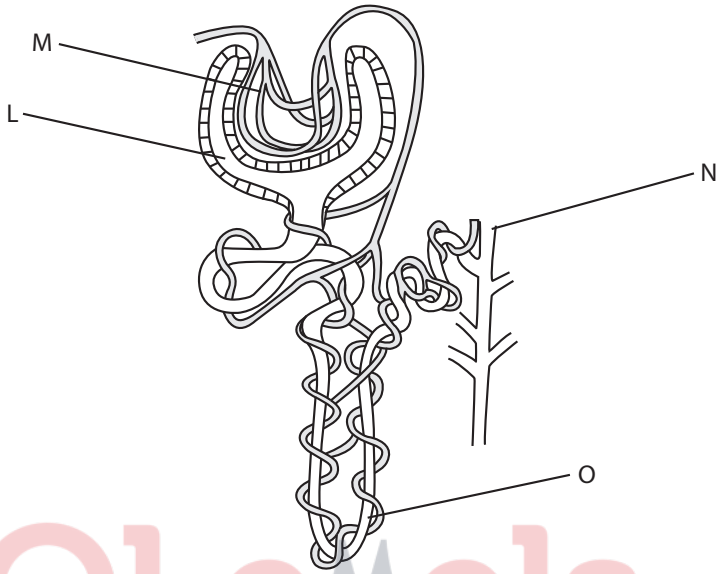
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7. Question 6 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O. **(4)**

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L. **(3)**

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7. Question 6 · part b.i · 3 marks

November 2023 Paper 1

Further practice · Attempt part b.i only.

(ii) Name **two** components of the blood that do not pass into L.

(2)

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

(1)

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

(5)

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(Total for Question 6 = 15 marks)

7. Question 6 · part b.i · 3 marks · Working and answer

Working and answer

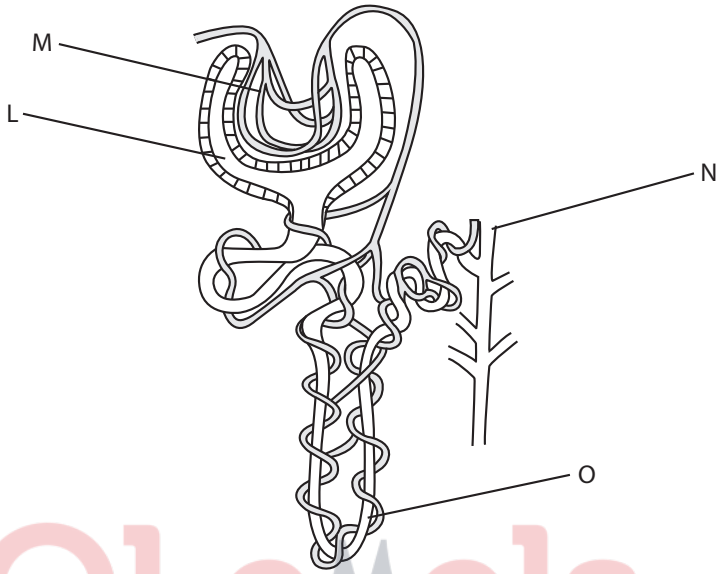
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8. Question 6 · part b.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O. **(4)**

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L. **(3)**

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8. Question 6 · part b.ii · 2 marks

November 2023 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Name **two** components of the blood that do not pass into L.

(2)

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

(1)

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

(5)

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(Total for Question 6 = 15 marks)

8. Question 6 · part b.ii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 5 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

5 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

Losartan belongs to a group of drugs known as angiotensin-II receptor antagonists. The body produces angiotensin-II, which is a substance that binds to receptors in blood vessels, causing them to narrow. This causes an increase in blood pressure. Losartan prevents angiotensin-II binding to these receptors, lowering blood pressure.

If high blood pressure is left untreated, a decrease in kidney function can result in proteinuria. This is a condition that causes high levels of protein to be found in a person's urine. People with high blood pressure and a resultant thickening of the left ventricle can be treated with Losartan to reduce the risk of stroke.

People with high blood pressure should not take Losartan if their liver is not functioning properly. Women who are pregnant or who are breast feeding should not take Losartan.

(a) Describe how the action of Losartan differs from the action of an ACE inhibitor. (lines 4 and 5)

(4)

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P 7 8 7 6 9 A 0 1 2 2 0

9. Question 5 · part c · 2 marks

June 2025 Paper 2

Further practice · Attempt part c only.

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(b) Explain how the structure of a named type of blood vessel allows angiotensin-II to increase blood pressure. (lines 2 to 5)

(4)

(c) Explain why proteinuria is a sign of problems with kidney function. (lines 6 to 8)

(2)

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(d) Suggest why high blood pressure causes the left ventricle to become thickened. (lines 8 to 9)

(2)

(e) Explain why people who have high blood pressure are advised not to take Losartan if their liver is not functioning properly. (lines 10 to 11)

(2)

(Total for Question 5 = 14 marks)

9. Question 5 · part c · 2 marks · Working and answer

Working and answer

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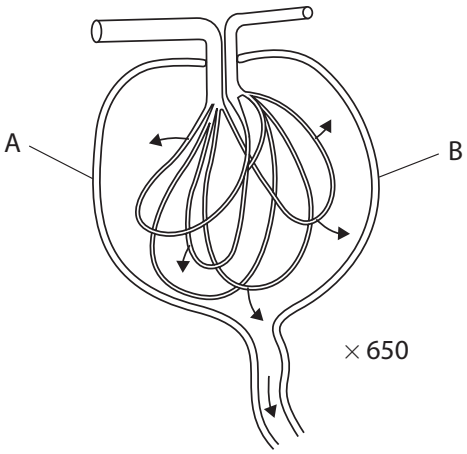
10. Question 1 · part a.iv · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iv only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

10. Question 1 · part a.iv · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iv only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
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1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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10. Question 1 · part a.iv · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 4 · part a.iii · 3 marks

June 2019 Paper 1R

Further practice · Attempt part a.iii only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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11. Question 4 · part a.iii · 3 marks

June 2019 Paper 1R

Further practice · Attempt part a.iii only.

(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

percentage = ... %

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11. Question 4 · part a.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 3 · part c.iii · 3 marks

June 2023 Paper 2

Further practice · Attempt part c.iii only.

3 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Diabetes is a very common condition that affects approximately five million people in the United Kingdom. Of these five million people, one person in four has the condition but is not aware that they have it. As a result, many people are at risk of high blood sugar levels.

There are two types of diabetes, Type 1 and Type 2. Type 2 diabetes is the cause of 90% of all cases of diabetes. This type is the result of the body not producing enough insulin or the body not reacting to the insulin.

Diabetes can cause damage to blood vessels and affect the level of fluid in the body. Some people with the condition develop problems with their eyesight and hearing.

Perhaps the most common symptoms of diabetes are passing more urine, a feeling of tiredness and wounds taking longer to heal than normal.

A quick blood test can be taken to show whether someone has the condition. Early diagnosis is important as diabetes increases the risks of some complications such as strokes and heart disease.

(a) Suggest why diabetes is described as a condition rather than a disease. (line 1)

(2)

.....

.....

.....

.....

(b) (i) Estimate how many people in the United Kingdom have diabetes without knowing that they have it. (lines 2 and 3)

(3)

number of people = ... million

(ii) Estimate how many people in the United Kingdom have Type 1 diabetes. (lines 5 and 6)

(2)

number of people = ... million

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P 7 2 5 8 7 A 0 6 2 4

12. Question 3 · part c.iii · 3 marks

June 2023 Paper 2

Further practice · Attempt part c.iii only.

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(c) (i) People with diabetes are at risk of high blood sugar levels.

State the name of the main sugar carried in blood. (line 4)

(1)

(ii) Suggest why some people with diabetes may develop problems with their eyesight and hearing. (lines 8 to 10)

(2)

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(iii) Explain why passing more urine is a symptom of diabetes. (line 11)

(3)

(Total for Question 3 = 13 marks)

12. Question 3 · part c.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 9 · part c · 4 marks

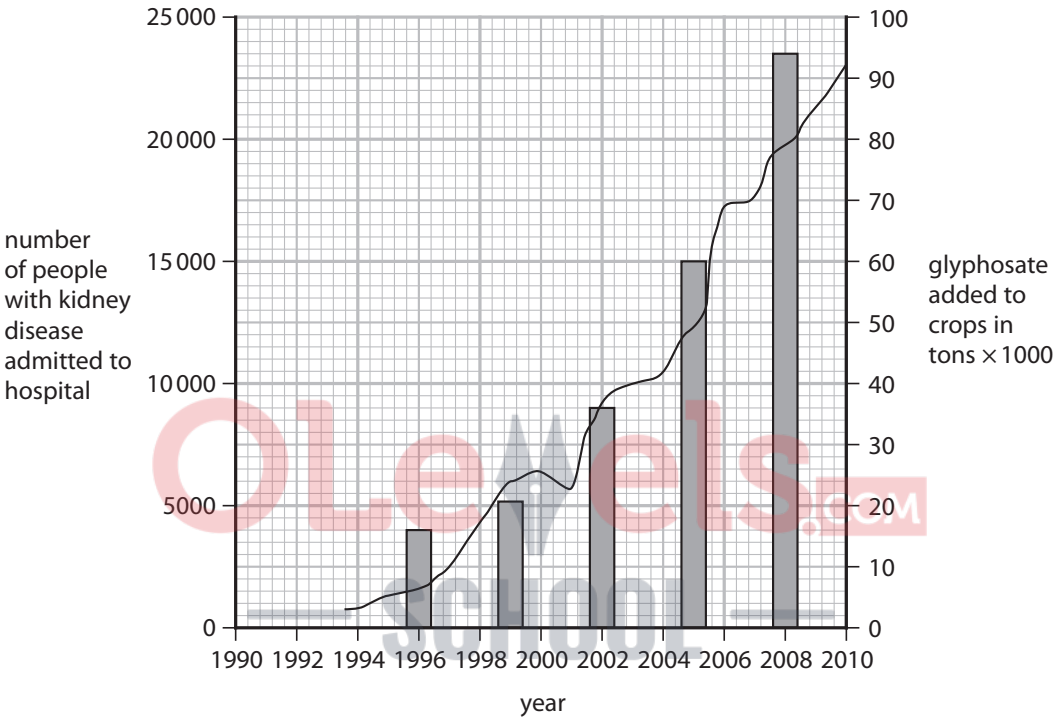
November 2020 Paper 1R

Further practice · Attempt part c only.

9 (a) Glyphosate is a chemical found in some weed killers.

The use of this chemical is linked to several health conditions including kidney disease.

The graph shows the number of people with kidney disease admitted to hospital and the amount of glyphosate added to crops.



(i) Calculate the percentage increase in the number of people admitted to hospital with kidney disease in 1996 compared with 2008.

(3)

percentage increase = ... %

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P 6 2 0 5 2 A 0 2 0 2 4

13. Question 9 · part c · 4 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

(ii) Give a reason why the graph does not confirm that glyphosate is a cause of kidney disease.

(1)

.....

.....

(b) Draw a diagram to show the structure of a kidney nephron.

Add these labels to the diagram.

- glomerulus
- proximal convoluted tubule
- Bowman's capsule

(4)

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13. Question 9 · part c · 4 marks

November 2020 Paper 1R

Further practice · Attempt part c only.

(c) There is evidence to show that glyphosate causes inflammation of the glomeruli and the kidney tubules.

Suggest how these symptoms will affect the normal functioning of the kidney and the waste product that it produces.

(4)

(Total for Question 9 = 12 marks)

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13. Question 9 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain how ADH changes urine volume and concentration

Hypothalamus detection, pituitary release of ADH, collecting duct permeability, and explaining urine differences on hot days, after drinking water or alcohol, or between samples.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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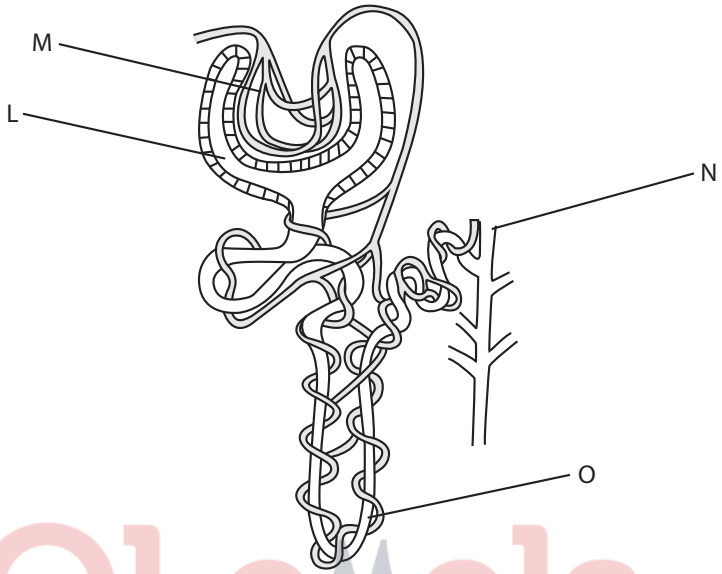
Type ID: HBI-ADH-OSMOREGULATION

1. Question 6 · part c · 5 marks

November 2023 Paper 1

Worked-example retry · Attempt part c only.

6 The diagram shows a kidney tubule.



(a) Name structures L, M, N, and O.

(4)

L .....

M .....

N .....

O .....

(b) (i) Describe the process that causes fluid to be formed in L.

(3)

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1. Question 6 · part c · 5 marks

November 2023 Paper 1

Worked-example retry · Attempt part c only.

(ii) Name **two** components of the blood that do not pass into L.

(2)

1 .....

2 .....

(iii) Label, with the letter G, the part of the kidney tubule where most of the glucose is reabsorbed.

(1)

(c) Explain why the volume of urine a person produces on a hot day is less than the volume of urine the same person produces on a cold day.

(5)

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(Total for Question 6 = 15 marks)

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1. Question 6 · part c · 5 marks · Working and answer

Working and answer

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2. Question 7 · part a.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.i only.

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7 The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

(1)

.....

(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

(3)

	Structure		
Function	**glomerulus**	**proximal tubule**	**renal vein**
regulation of water level			
ultrafiltration			
reabsorption of glucose			
removal of blood from kidney			
supply of blood to kidney			
removal of urine from collecting duct			

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2. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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3. Question 6 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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3. Question 6 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

OLevels.com SCHOOL

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

3. Question 6 · part c · 2 marks

November 2020 Paper 2

Further practice · Attempt part c only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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3. Question 6 · part c · 2 marks · Working and answer

Working and answer

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4. Question 5 · part c · 4 marks

June 2022 Paper 1

Further practice · Attempt part c only.

5 (a) There is a difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Design an investigation to determine the difference in the volume of urine produced on a hot day and the volume of urine produced on a cold day.

Include experimental details in your answer and write in full sentences.

(4)

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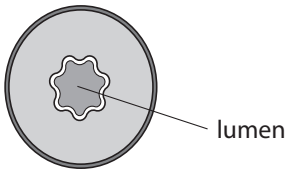
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4. Question 5 · part c · 4 marks

June 2022 Paper 1

Further practice · Attempt part c only.

(b) The diagram shows a cross-section of a blood vessel in the skin on a cold day.



Explain the changes in this blood vessel on a hot day.

(3)

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(c) Describe how ADH regulates the volume of water in the body on a hot day.

(4)

(Total for Question 5 = 11 marks)

4. Question 5 · part c · 4 marks · Working and answer

Working and answer

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5. Question 7 · part b.ii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

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7 The renal system is involved in osmoregulation.

(a) (i) Name the area of the brain that detects the water content of the blood.

(1)

.....

(ii) The kidneys are part of the renal system.

Put one tick (✓) in each column of the table to identify the function of different structures in the kidney.

(3)

	Structure		
Function	**glomerulus**	**proximal tubule**	**renal vein**
regulation of water level			
ultrafiltration			
reabsorption of glucose			
removal of blood from kidney			
supply of blood to kidney			
removal of urine from collecting duct			

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5. Question 7 · part b.ii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

(b) The table shows the percentage composition of two urine samples taken from one person.

Not all of the components are shown.

The first sample was taken at midday and the second sample was taken four hours later.

Component	Percentage composition	
	first sample	second sample
ammonia	0.05	0.08
uric acid	0.03	0.08
ions	1.34	1.46
urea	2.00	2.80
water	95.00	91.50

(i) Describe how urea is produced in the body.

(2)

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5. Question 7 · part b.ii · 5 marks

January 2023 Paper 2

Further practice · Attempt part b.ii only.

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(ii) The ADH response changes the percentage composition of urine.

Explain the ADH response that occurred in the person's body between the first and the second sample being collected.

(5)

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(Total for Question 7 = 11 marks)

5. Question 7 · part b.ii · 5 marks · Working and answer

Working and answer

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6. Question 9 · part whole · 5 marks

June 2024 Paper 1R

Further practice · Attempt part whole only.

9 Explain the role of ADH in osmoregulation.

(5)

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(Total for Question 9 = 5 marks)

6. Question 9 · part whole · 5 marks · Working and answer

Working and answer

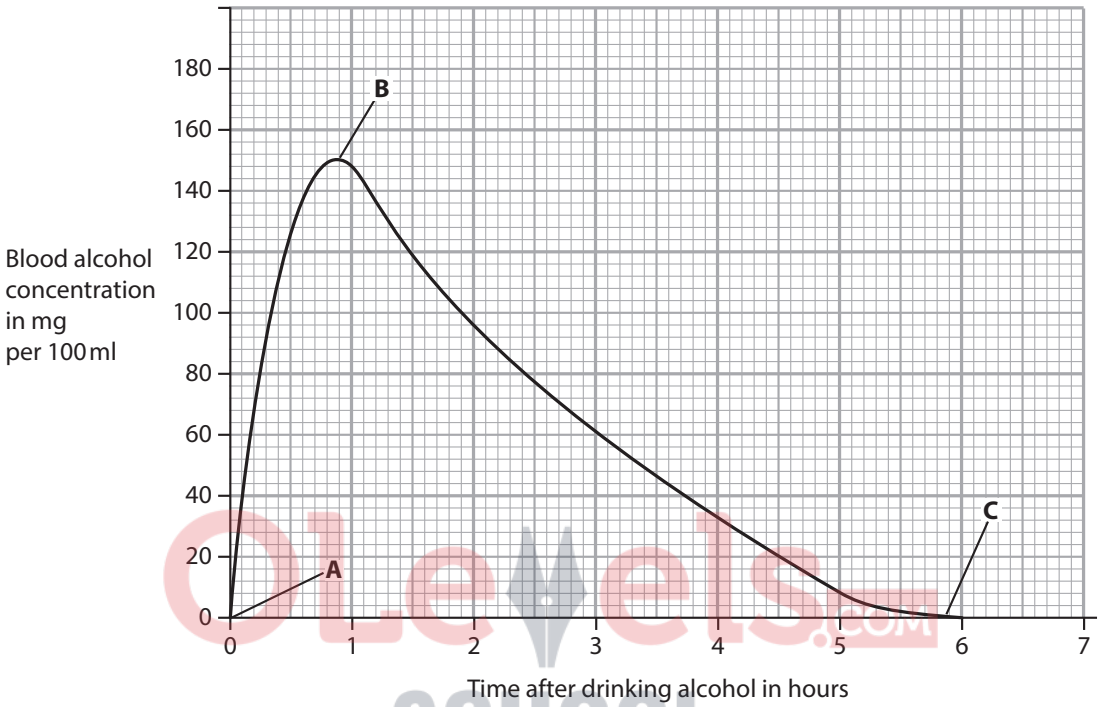
OLevels.com SCHOOL

7. Question 6 · part c · 5 marks

June 2025 Paper 1

Further practice · Attempt part c only.

6 The graph shows the changes in blood alcohol concentration of a person who drinks alcohol.



(a) (i) Explain the shape of the graph between points A and B.

(2)

(ii) Explain the shape of the graph between points B and C.

(3)

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7. Question 6 · part c · 5 marks

June 2025 Paper 1

Further practice · Attempt part c only.

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(b) In the United Kingdom the legal limit to drive is less than 80 mg per 100 ml of alcohol in the blood.

(i) The person shown in the graph drinks alcohol and the alcohol level in the blood goes above the legal limit to drive.

Determine how many hours after drinking alcohol it would be legal for the person shown in the graph to drive.

(1)

(ii) Explain why it is against the law to drive with high levels of alcohol in the blood.

(4)

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7. Question 6 · part c · 5 marks

June 2025 Paper 1

Further practice · Attempt part c only.

(c) Alcohol reduces the production of ADH.

Explain the effect on the urine produced by a person drinking alcohol.

(5)

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(Total for Question 6 = 15 marks)

7. Question 6 · part c · 5 marks · Working and answer

Working and answer

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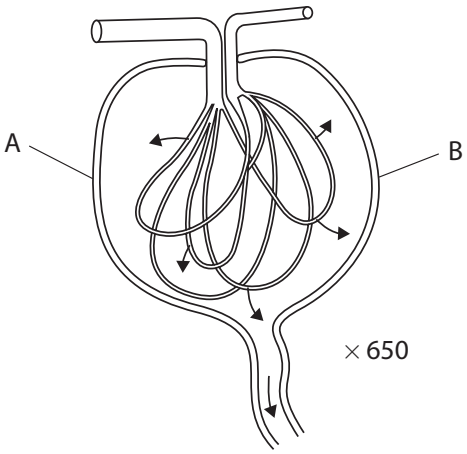
8. Question 1 · part b.iii · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.iii only.

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1 (a) The diagram shows the Bowman's capsule and glomerulus of a kidney tubule (nephron).



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(i) Complete the diagram to show the missing parts of the kidney tubule.

(3)

(ii) Calculate the actual width of the Bowman's capsule, in mm, from the point labelled A to the point labelled B.

(3)

width of Bowman's capsule = ... mm

8. Question 1 · part b.iii · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.iii only.

(iii) Name two components of the blood that would not pass into the Bowman's capsule.

(2)

1 .....

2 .....

(iv) Name one component of the glomerular filtrate that is completely reabsorbed into the blood from the tubule.

(1)

.....

(b) A scientist uses this method to investigate the effect of room temperature on urine production.

On day 1, a person drinks 1 dm^3 of water. Urine samples from the person are collected at intervals during a time period of four hours.

The room temperature is 20 °C.

This is then repeated on day 2, when the room temperature is 30 °C.

The table shows the results.

Time after drinking in hours	Volume of urine in cm^3	
	day 1 (20 °C)	day 2 (30 °C)
0.5	70	50
1.5	175	70
2.0	300	200
2.5	250	190
3.0	210	150
4.0	100	50

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8. Question 1 · part b.iii · 3 marks

January 2022 Paper 2

Further practice · Attempt part b.iii only.

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(i) State two factors that should be controlled in the investigation.

(2)

1 .....

.....

2 .....

.....

(ii) The mean rate of urine production on day 1 is 276 cm^3 per hour.

Calculate the mean rate of urine production on day 2 during the 4 hour time period.

(2)

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mean rate of urine production on day 2 = ... cm^3 per hour

(iii) Explain the difference between the mean rate on day 1 and the mean rate on day 2.

(3)

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(Total for Question 1 = 16 marks)

8. Question 1 · part b.iii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 6 · part e · 3 marks

November 2020 Paper 2

Further practice · Attempt part e only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The effects of alcohol

Humans have been drinking alcohol for thousands of years. Drinking too much alcohol over many years can cause diabetes, malnutrition and diseases of the central nervous system and the liver. One immediate side effect of drinking too much alcohol is increased urination.

Beer is about 95% water and only 5% alcohol. The liver converts that 5% of alcohol into the same volume of water and some carbon dioxide. So if a person drinks 200 cm^3 of beer (one unit), the end result is 200 cm^3 of water. However, this person will not just urinate 200 cm^3 of urine, in fact the person urinates a total of 320 cm^3 of urine. So in general each unit of alcohol makes the person urinate an extra 120 cm^3 of urine more than the normal urine output.

A person with a mass of 60 kg produces 60 cm^3 of urine an hour and a person with a mass of 80 kg produces 80 cm^3 an hour and so on. Therefore if a 60 kg person drinks 200 cm^3 of beer in an hour they will produce 60 cm^3 of urine plus 320 cm^3, making a total of 380 cm^3.

Alcohol interferes with the mechanism that regulates water levels in our body because alcohol affects a small gland called the pituitary gland. One hormone released by the pituitary gland is anti-diuretic hormone (ADH). Alcohol reduces how much ADH is released by the pituitary. Each unit of alcohol that a person drinks forces the kidneys to produce an extra 120 cm^3 of urine more than the normal output.

Logic suggests that drinking lots of water would replace the 120 cm^3 of urine produced, but it is not that simple. The body only retains half the water that a person drinks. The other half leaves the body in urine, so the person will become dehydrated even if they drink lots of water.

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9. Question 6 · part e · 3 marks

November 2020 Paper 2

Further practice · Attempt part e only.

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(a) (i) Where in the body is the pituitary gland (line 16)?

(1)

- ☒ **A** brain
- ☒ **B** kidney
- ☒ **C** liver
- ☒ **D** pancreas

(ii) Which system includes the pituitary gland (line 16)?

(1)

- ☒ **A** digestive
- ☒ **B** endocrine
- ☒ **C** nervous
- ☒ **D** skeletal

(b) A person weighs 75 kg and drinks 400 cm^3 of beer in an hour (lines 11–14).

(i) Calculate the volume of urine produced by the person in this hour.

(3)

volume of urine = ... cm^3

(ii) Calculate the percentage increase in the volume of urine produced by this person in this hour.

(3)

percentage increase = ...%

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9. Question 6 · part e · 3 marks

November 2020 Paper 2

Further practice · Attempt part e only.

(c) Describe the effect of ADH on the kidney (line 17).

(2)

(d) Suggest the effects of dehydration on the blood after increased urination.

(2)

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(e) Explain why half of the extra water that the person drinks still leaves the body in the urine (lines 21–24).

(3)

(Total for Question 6 = 15 marks)

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9. Question 6 · part e · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Explain diffusion across a partially permeable membrane in dialysis and Visking tubing

How dialysis fluid removes urea but keeps glucose and salts, why glucose is added, and Visking tubing models where small molecules pass and starch stays.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-PARTIALLY-PERMEABLE-MEMBRANE-DIALYSIS

1. Question 8 · part b.i · 4 marks

January 2020 Paper 1

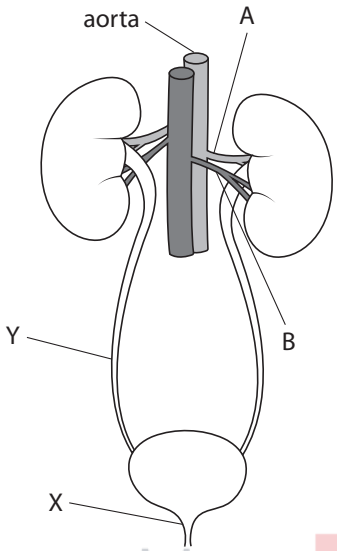
Worked-example retry · Attempt part b.i only.

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8 The diagram shows part of the human renal system.



(a) (i) Give the names of structures X and Y.

(2)

structure X .....

structure Y .....

(ii) Describe the function of the aorta.

(2)

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(iii) Compare the composition of the blood in vessels A and B.

(2)

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1. Question 8 · part b.i · 4 marks

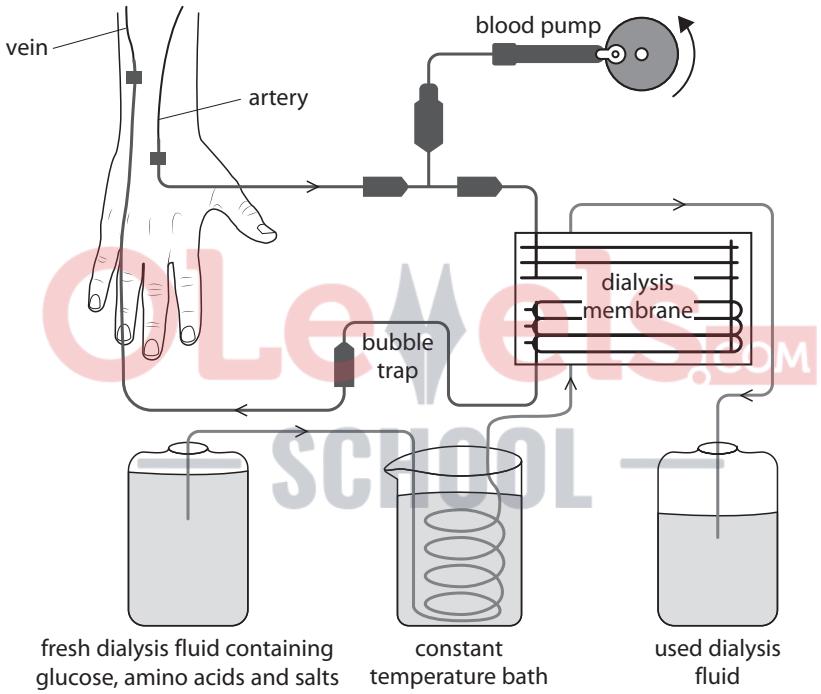
January 2020 Paper 1

Worked-example retry · Attempt part b.i only.

(b) Dialysis is a treatment used for people with kidney failure.

Blood containing waste substances is pumped through a dialysis machine. The dialysis machine filters the waste from the blood. The cleaned blood is then returned to the patient.

The diagram shows how a dialysis machine filters the blood of a person with kidney failure.



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1. Question 8 · part b.i · 4 marks

January 2020 Paper 1

Worked-example retry · Attempt part b.i only.

(i) Explain how diffusion removes waste substances from the patient's blood inside the dialysis machine.

(4)

(ii) A chemical is added to the patient's blood as it enters the dialysis machine. This chemical prevents the blood from clotting.

Explain why it is important to add this chemical as the blood enters the machine, rather than just before the blood returns to the body.

(3)

(Total for Question 8 = 13 marks)

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1. Question 8 · part b.i · 4 marks · Working and answer

Working and answer

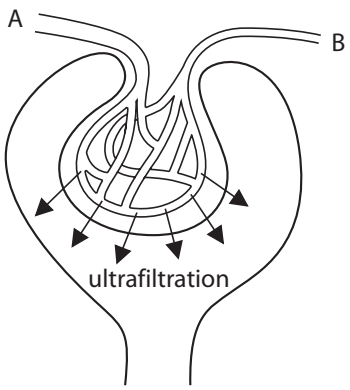
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2. Question 6 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

6 (a) The diagram shows a Bowman's capsule and its associated blood supply.



(i) Describe what is meant by the term **ultrafiltration**.

(2)

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(ii) State three differences between the composition of blood in vessel A and the composition of blood in vessel B.

(3)

1 .....

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2 .....

.....

3 .....

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2. Question 6 · part b · 3 marks

June 2021 Paper 1

Further practice · Attempt part b only.

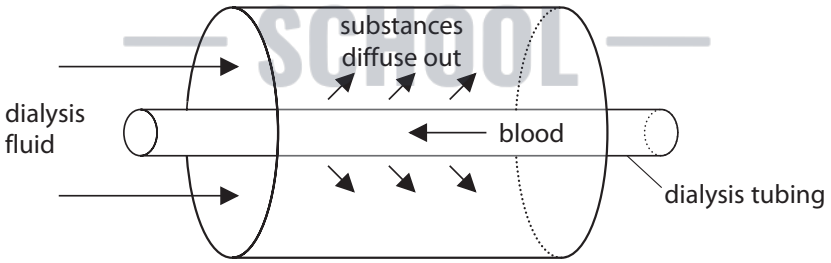
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(iii) Explain how blood vessel A and blood vessel B help the process of ultrafiltration.

(4)

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(b) The diagram shows part of a kidney dialysis machine.



The dialysis fluid contains glucose at the same concentration as in the blood flowing through the dialysis tubing.

Suggest the effect on a person undergoing dialysis if the concentration of glucose was below that of the blood flowing through the dialysis tubing.

(3)

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(Total for Question 6 = 12 marks)

2. Question 6 · part b · 3 marks · Working and answer

Working and answer

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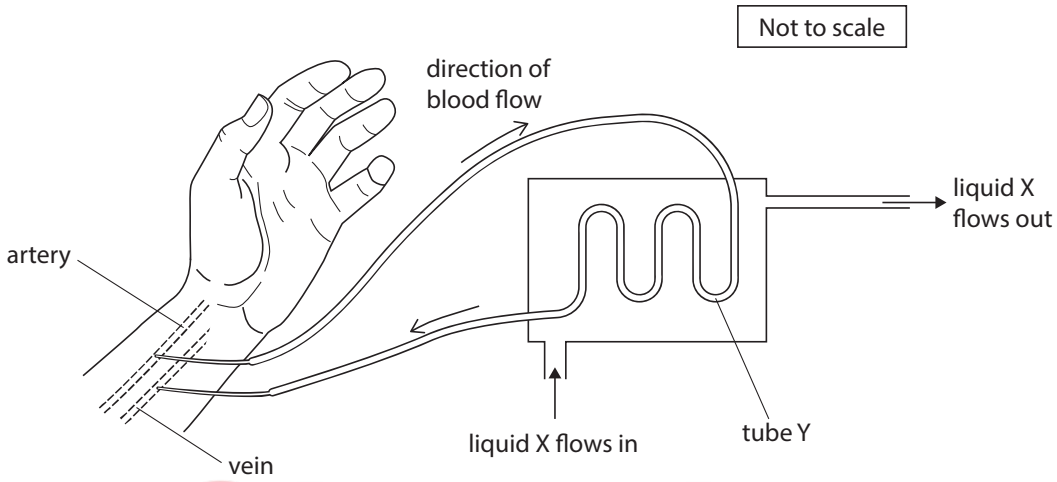
3. Question 5 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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P 7 4 6 2 8 A 0 1 2 2 0

3. Question 5 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

(2)

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

(1)

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(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

(3)

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3. Question 5 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

(c) Explain why people who have to use kidney dialysis machines are often told to eat less protein in their diet.

(4)

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(Total for Question 5 = 15 marks)

Working and answer

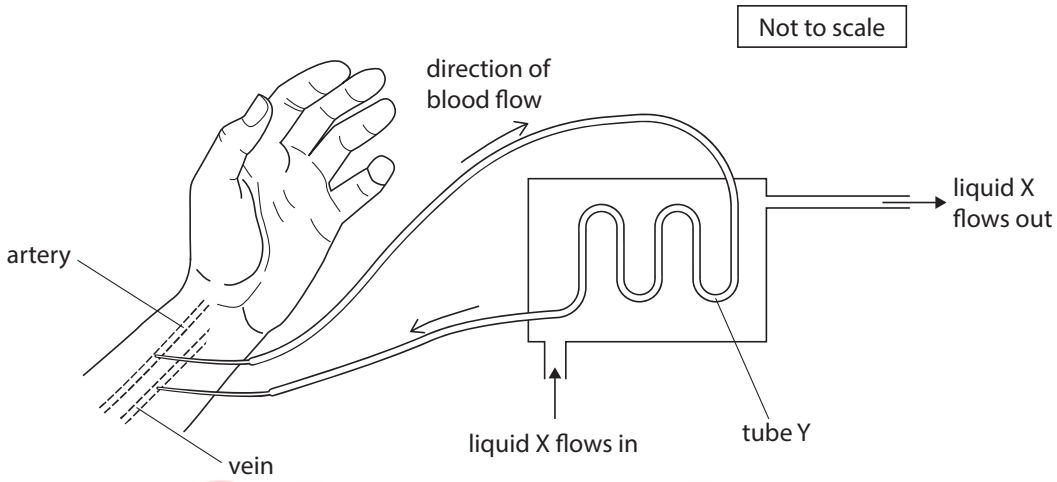
4. Question 5 · part a.iv · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.iv only.

5 The diagram shows a kidney dialysis machine.

The wall of tube Y is partially permeable so it allows the passage of small molecules.



(a) (i) Explain why the blood flow is in the direction shown in the diagram.

(4)

(ii) State which part of the kidney nephron has the same role as tube Y.

(1)

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4. Question 5 · part a.iv · 1 mark

June 2025 Paper 1R

Further practice · Attempt part a.iv only.

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(iii) Explain why tube Y has a length of several metres in the dialysis machine.

(2)

(iv) State one difference between the composition of liquid X as it flows into the machine and as it flows out of the machine.

(1)

(b) Glucose is added to liquid X before it passes into the dialysis machine.

Explain why glucose is added.

(3)

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4. Question 5 · part a.iv · 1 mark · Working and answer

Working and answer

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5. Question 2 · part b · 5 marks

January 2022 Paper 1

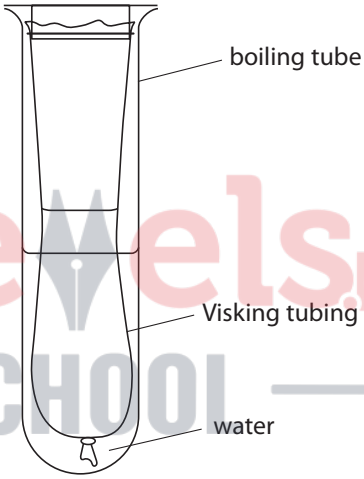
Further practice · Attempt part b only.

2 A student investigates absorption in the ileum by using a model gut made of Visking tubing.

This is the student's method.

- tie a knot in one end of a 15 cm length of Visking tubing to make a bag
- place 5 cm^3 of starch suspension and 5 cm^3 of glucose solution into the bag
- rinse the outside of the bag with water
- place the bag into a boiling tube containing distilled water
- leave the bag for one hour

The diagram shows the student's apparatus.



After an hour the contents of the Visking tubing bag are tested for glucose and starch.

The water in the boiling tube is also tested for glucose and starch.

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5. Question 2 · part b · 5 marks

January 2022 Paper 1

Further practice · Attempt part b only.

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(a) (i) Describe how the student should test the contents of the Visking tubing bag for glucose.

(4)

(ii) Describe how the student should test the contents of the Visking tubing bag for starch.

(3)

(iii) Explain why the outside of the Visking tubing bag was rinsed with water before being placed in the boiling tube.

(3)

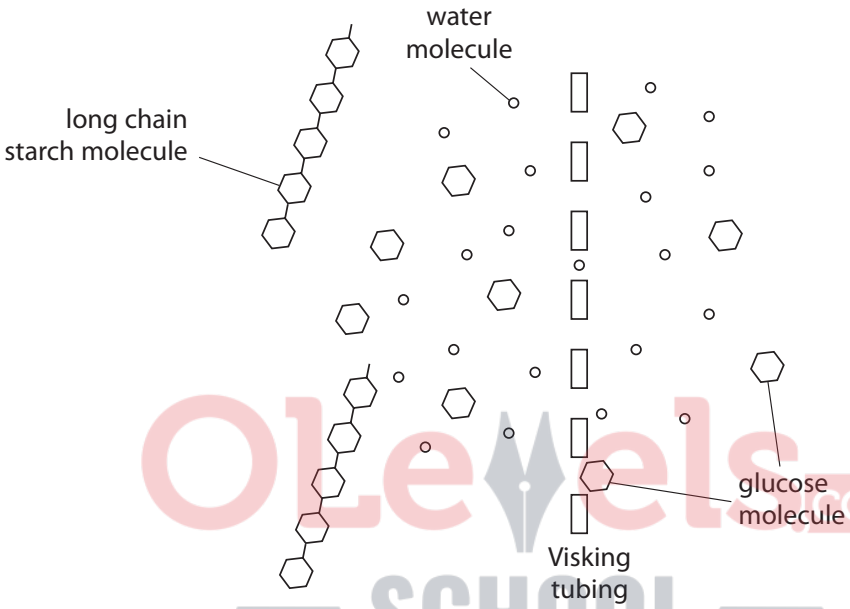
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5. Question 2 · part b · 5 marks

January 2022 Paper 1

Further practice · Attempt part b only.

(b) The student draws this diagram to explain the results obtained.



Explain the results the student obtains.

Refer to the diagram in your answer.

(5)

(Total for Question 2 = 15 marks)

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5. Question 2 · part b · 5 marks · Working and answer

Working and answer

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6. Question 5 · part b.iii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

5 Glucose is a product of digestion.

Glucose moves from the small intestine into the blood by diffusion.

(a) Explain why the rate of diffusion of glucose into the blood stream is affected by the concentration of glucose in the small intestine.

(3)

(b) A student uses different concentrations of glucose solution to investigate the rate of diffusion from the small intestine into the blood.

The list gives some of the equipment that the student uses in the investigation.

- Visking tubing (to represent the small intestine)
- boiling tube
- different concentrations of glucose solutions
- stop clock

(i) Name the chemical the student uses to test for glucose.

(1)

(ii) Name one other piece of equipment that the student would need to use to test for glucose.

(1)

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6. Question 5 · part b.iii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

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(iii) Draw a diagram to show how the student could use the equipment to investigate the effect of glucose concentration on the rate of diffusion.

You should add these labels to your diagram.

- Visking tubing
- boiling tube
- sugar solution
- water

(3)

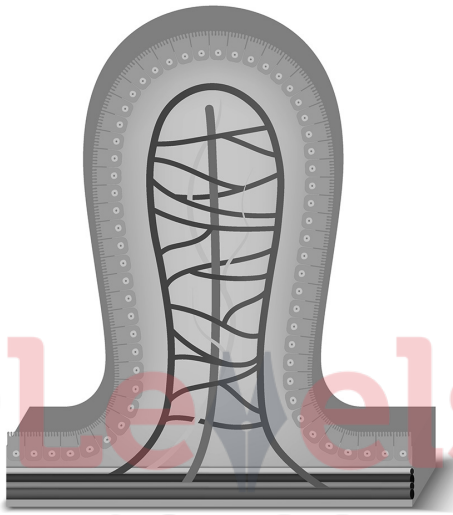
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6. Question 5 · part b.iii · 3 marks

June 2025 Paper 2R

Further practice · Attempt part b.iii only.

(c) The diagram shows a structure found in the small intestine. This structure increases the rate of diffusion of molecules such as glucose from the small intestine into the blood stream.



(Source: © Dee-sign / Shutterstock)

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(i) Name this structure.

(1)

.....

(ii) Explain how these structures increase the rate of diffusion of glucose from the small intestine into the blood stream.

(3)

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(Total for Question 5 = 12 marks)

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6. Question 5 · part b.iii · 3 marks · Working and answer

Working and answer

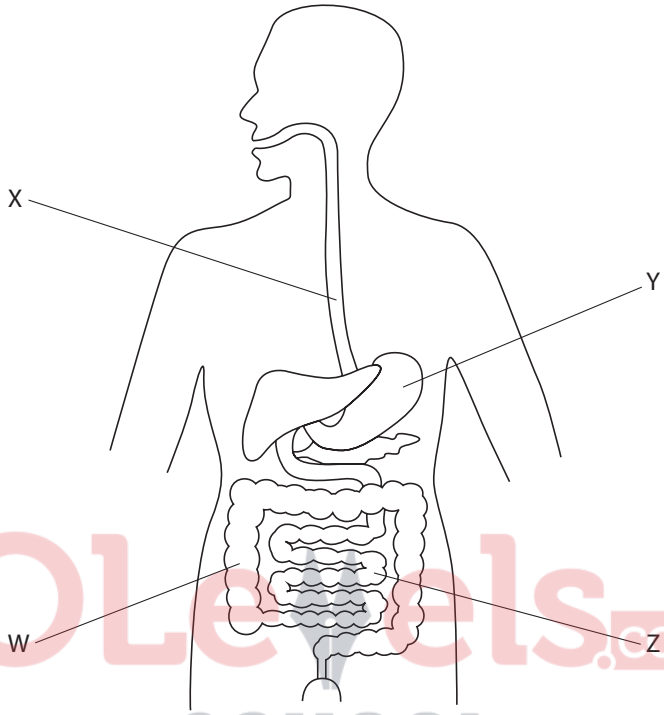
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7. Question 2 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

2 (a) The diagram shows the human digestive system.



(i) Name the structures labelled W, X, Y and Z.

(4)

W .....

X .....

Y .....

Z .....

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7. Question 2 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

relax	coordination	contract	liquid
peristalsis	muscles	enzymes	bolus

Use words from the box to complete the passage.

(4)

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

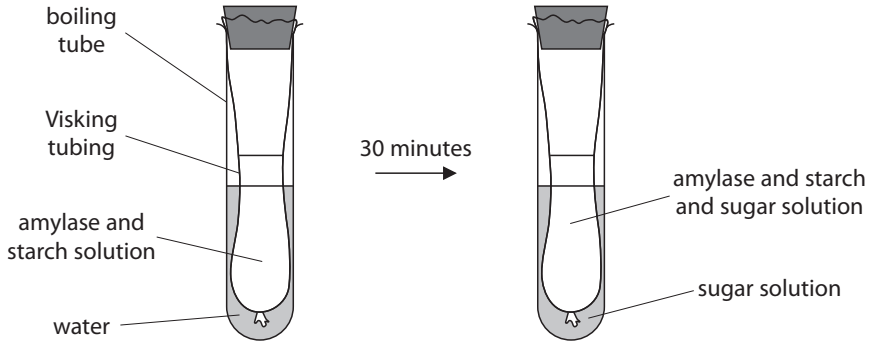
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7. Question 2 · part b.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.ii only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.



(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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P 7 0 9 4 9 A 0 8 2 4

7. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

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